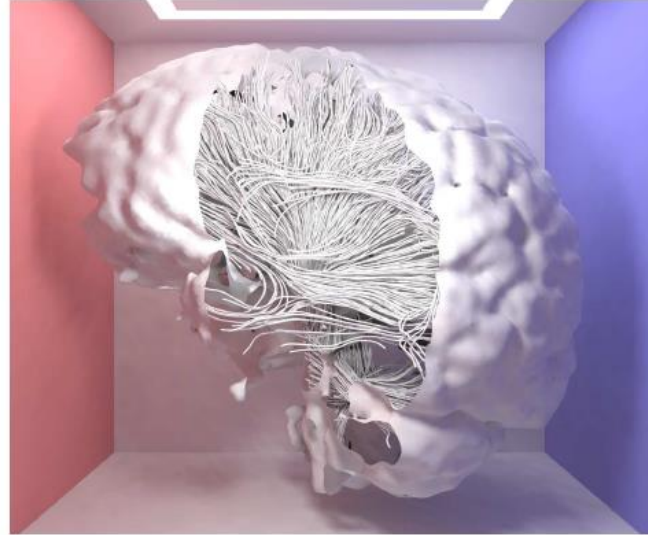




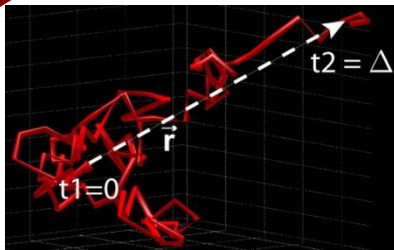
MRI Processing Theory: Diffusion Weighted Imaging

Yasser Alemán-Gómez

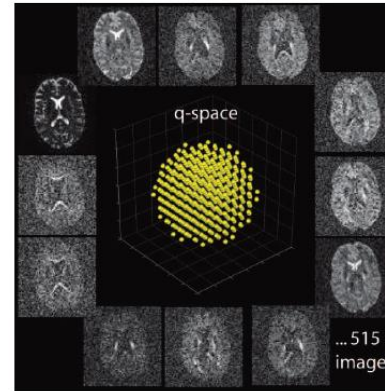
Radiology department. CHUV, Switzerland
October 1st, 2026

Index

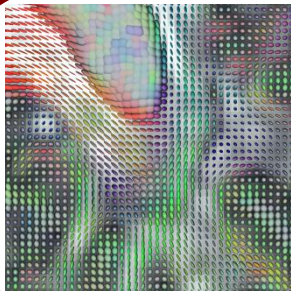
DWMRI: Physical Basis



Diffusion Weighted Magnetic Resonance Imaging (DWMRI)



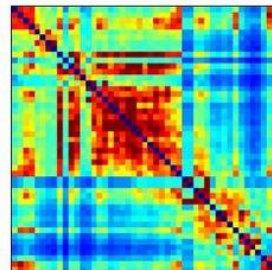
Fiber Reconstruction Algorithms

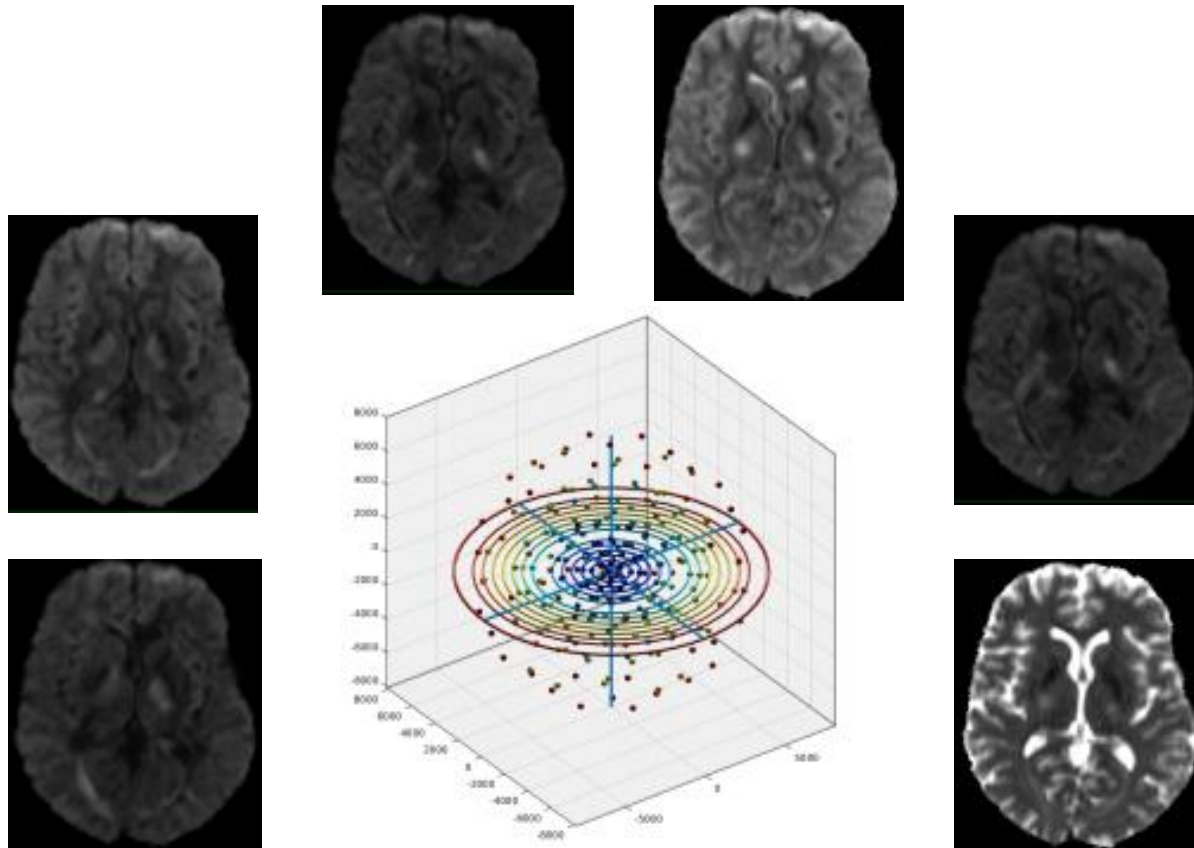


Fiber Tracking



Anatomical Connectivity





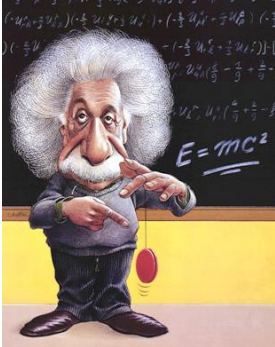
Diffusion-Weighted MRI: Physical Basis

Brownian Movement



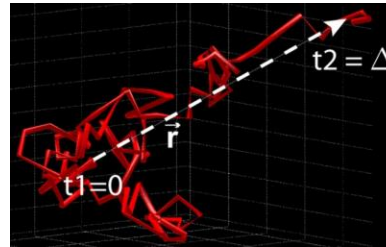
Robert Brown (1773-1858) (*Scottish Botanist*)

1827: Examined pollen grains suspended in water under a microscope, observed the pollen grains executing a **continuous jittery motion**



Albert Einstein (1879-1955) (*Physicist*)

- Brownian motion as responsible, at a microscopic scale, for the diffusion phenomenon at a macroscopic scale
- Mean Square Displacement depends on:
 - ❑ Fluid Temperature
 - ❑ Fluid Viscosity
 - ❑ Particle size

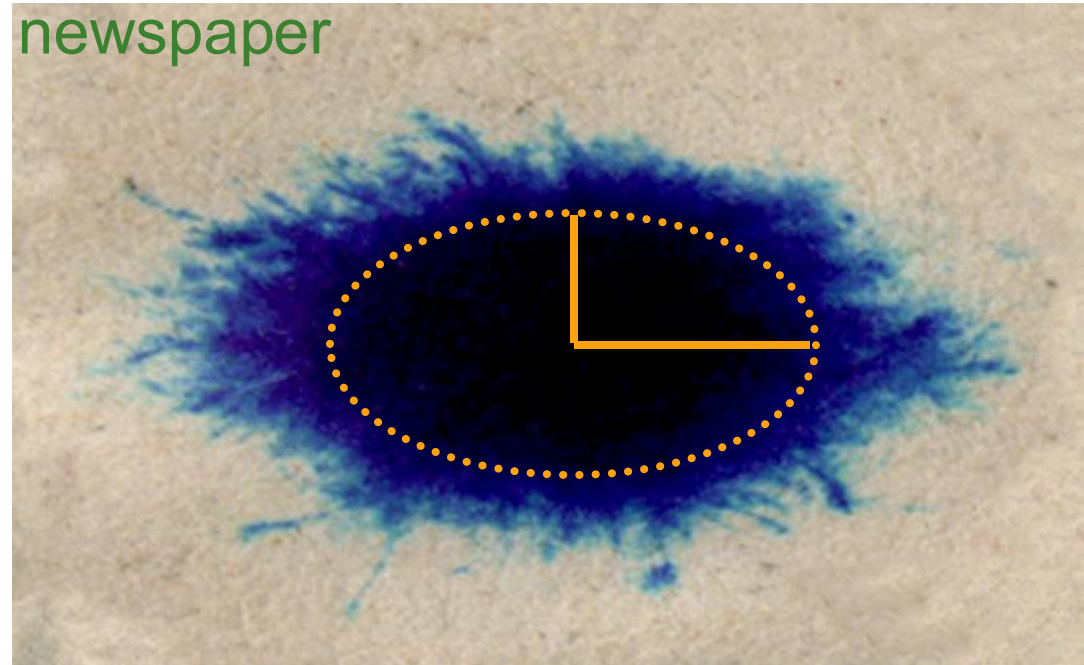


$$\langle \Delta r^2 \rangle = 6D\tau$$

Diffusion coefficient

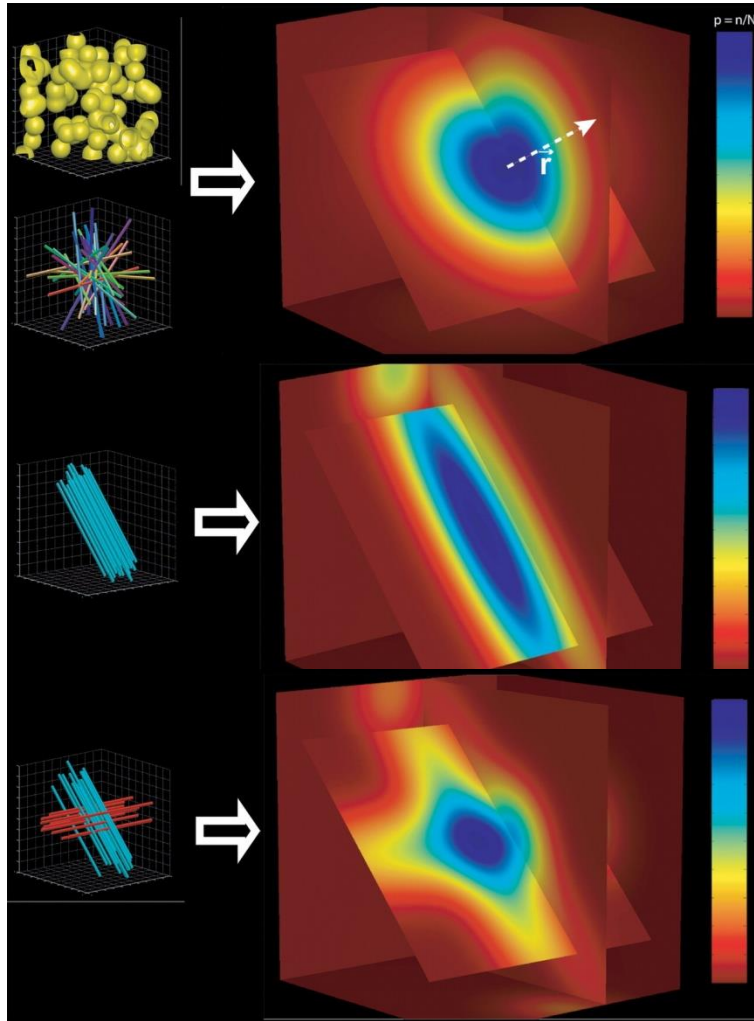
Isotropic and anisotropic diffusion

- **Isotropic Diffusion:** *Diffusion is the same in all directions, e.g. a pure liquid.*
- **Anisotropic Diffusion:** *Diffusion is not the same in all directions because the medium does not have the same properties (e.g. viscosity, permeability) in all directions.*

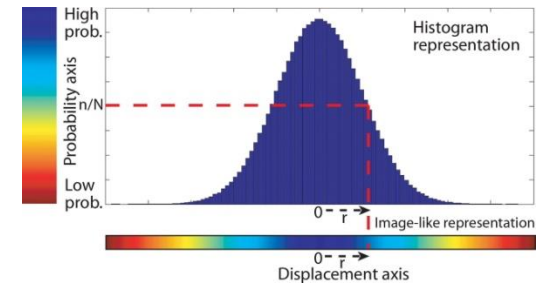


Molecular Displacement (Probability Displacement Function)

Diffusion profiles within a voxel



Free Diffusion. Gaussian Distribution



$$P(\vec{r}): N(0, \Sigma)$$

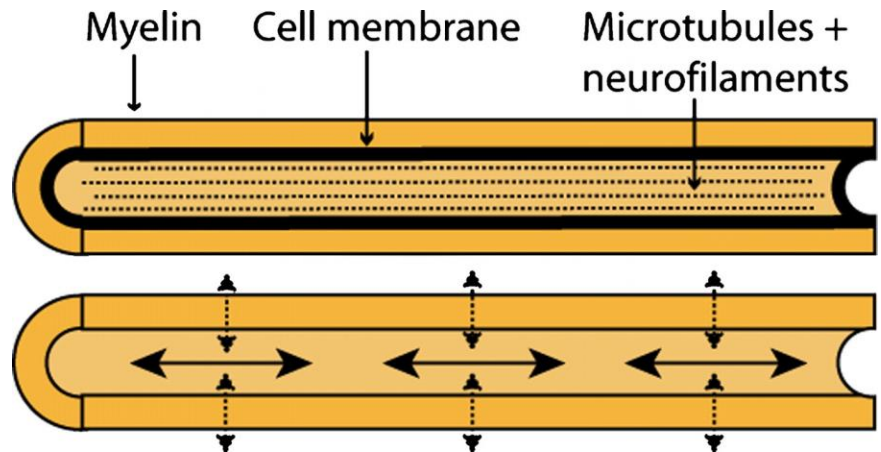
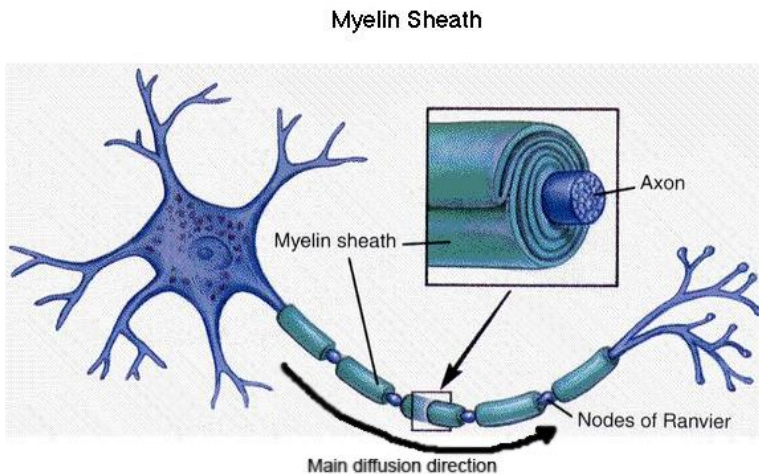
Non-Gaussian Distribution

Diffusion depends on the cyto-architectonical structure of tissues that constrain the diffusion process

$$P(\vec{r}): ???$$

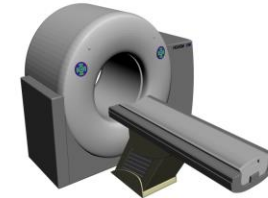
Diffusion in Brain Tissue: Some Facts

- Diffusion in biological tissues is often anisotropic
- White Matter:
 - Axonal membranes: Responsible for diffusion anisotropy
 - Myelin and axons bundles: Diffusion anisotropy modulators (Beaulieu and Allen, et al 1994)
 - Diffusion along a fiber bundle is the main diffusion direction



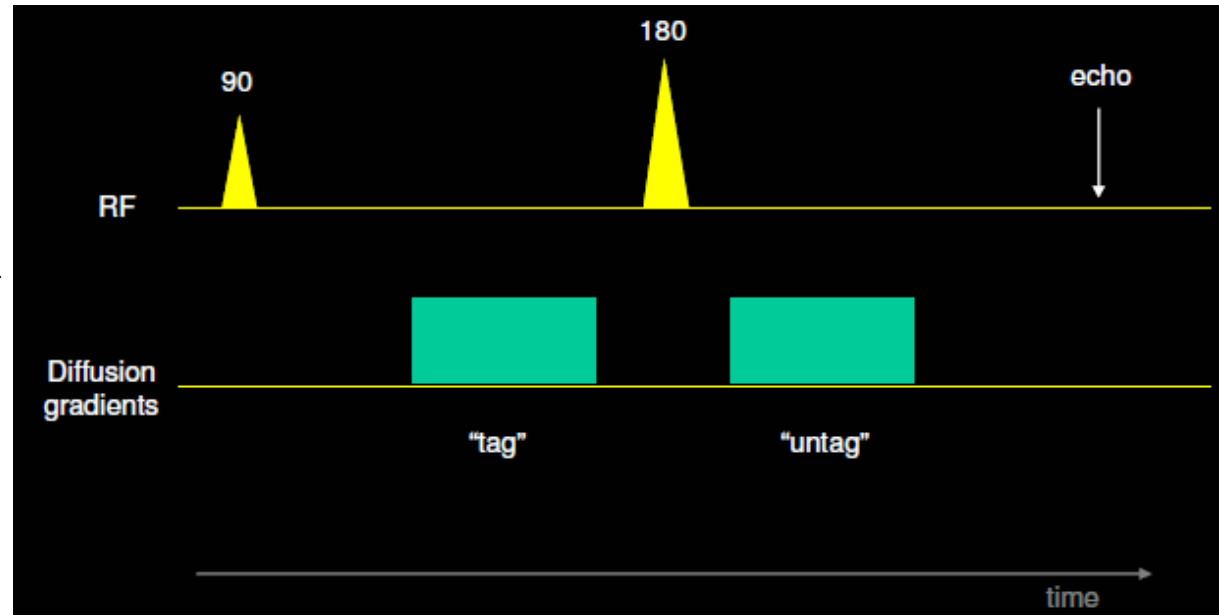
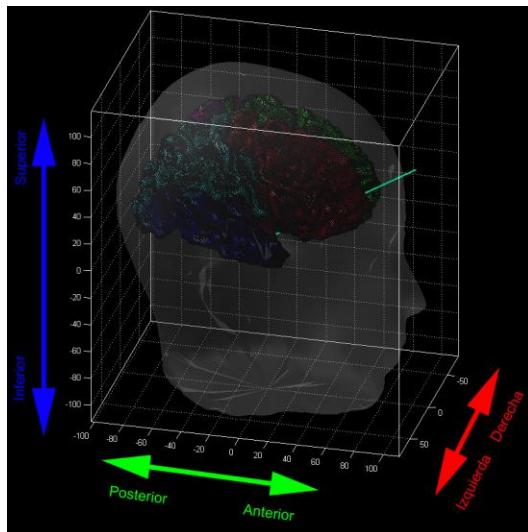
Pulse sequence

MR Scanner



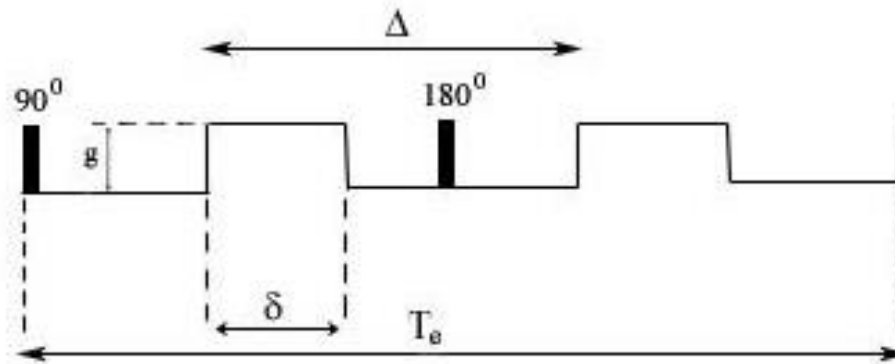
- How to measure Diffusion in an MR scanner
 1. 'Label' molecules according to their position
 2. Allow time for molecules to diffuse
 3. 'Un-label' by reversing operation 1

Spin Echo Pulse Sequence



Pulse sequence: Signal Equation

Stejskal-Tanner sequence



- δ Diffusion-sensitizing gradient duration
- Δ Distance between pulses
- $\vec{g}$ Diffusion-sensitizing gradient vector
- T_e Echo Time

Diffusion Magnetic Resonance Signal obtained after the application of a magnetic gradient over the direction $\vec{g}$

$$S = S_0 e^{-bD}$$

$\rightarrow$ Apparent diffusion coefficient (ADC)
 $\rightarrow$ Diffusion Magnetic Resonance Signal without magnetic pulse gradient

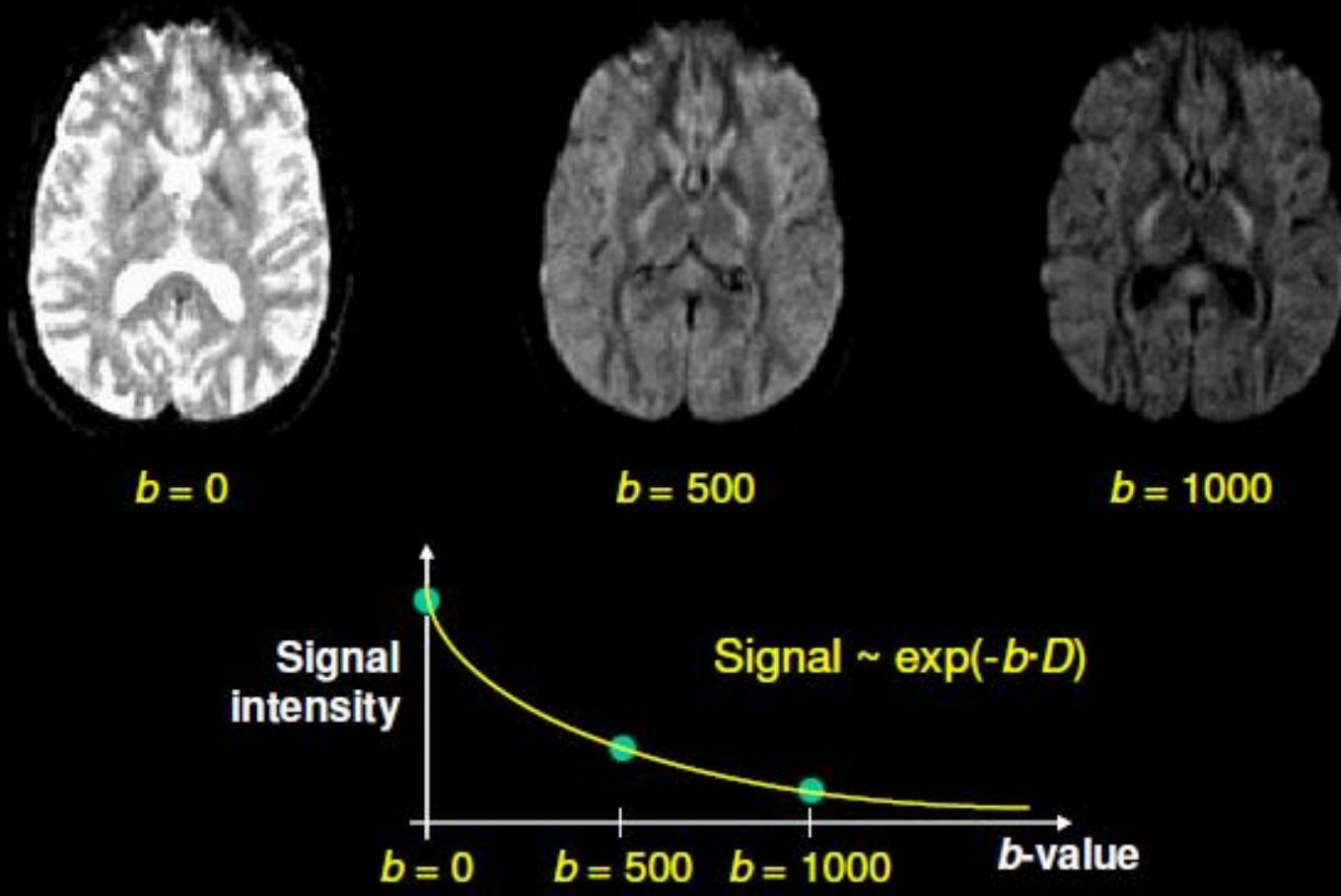
Where:

$$b = \gamma^2 \delta^2 \left| \frac{\vec{r}}{g} \right|^2 (\Delta - \delta/3)$$

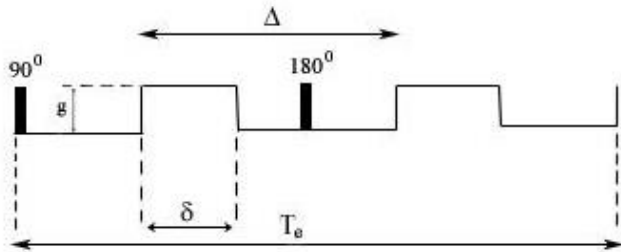
Scalar that depends on the pulse sequence

B-Value

b-value reflects strength of applied diffusion-weighting



B-Value



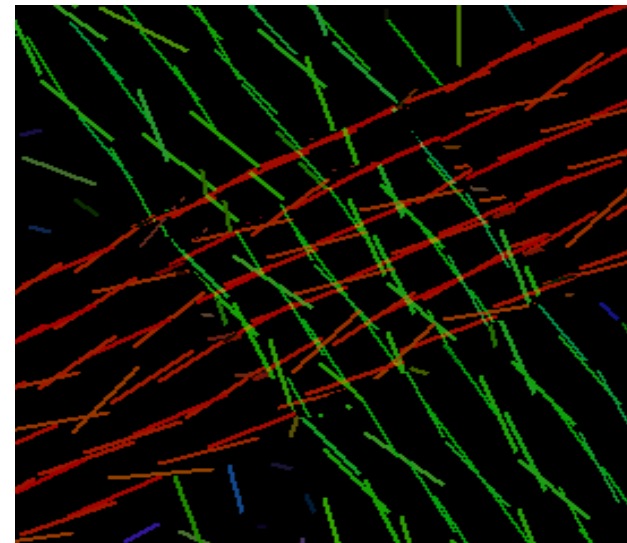
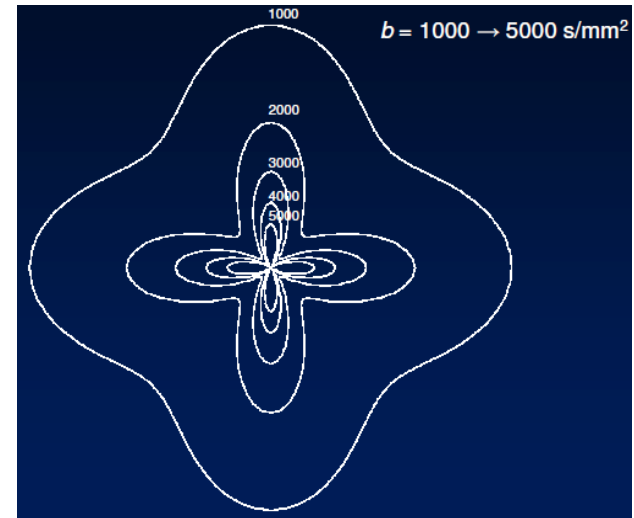
δ Diffusion-sensitizing gradient duration

Δ Distance between pulses

$\vec{g}$ Diffusion-sensitizing gradient vector

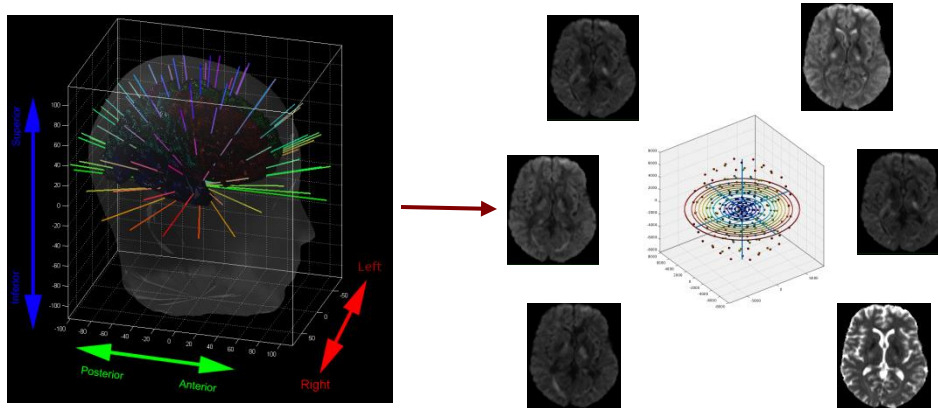
T_e Echo Time

$$b = \gamma^2 \delta^2 |\vec{g}|^2 (\Delta - \delta/3)$$

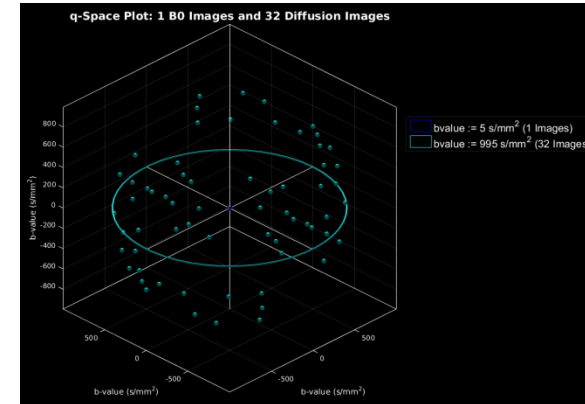


Acquisition schemes

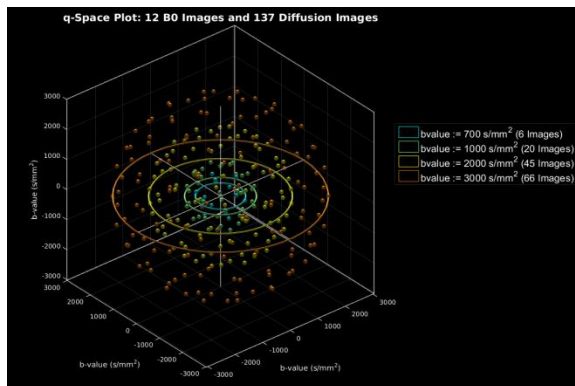
Acquisition scheme



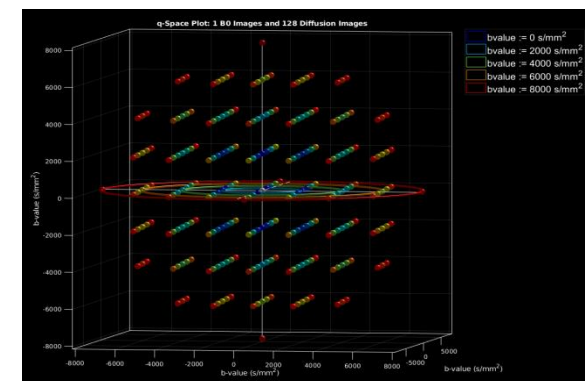
DTI acquisition



HARDI Multi shell acquisition



DSI acquisition



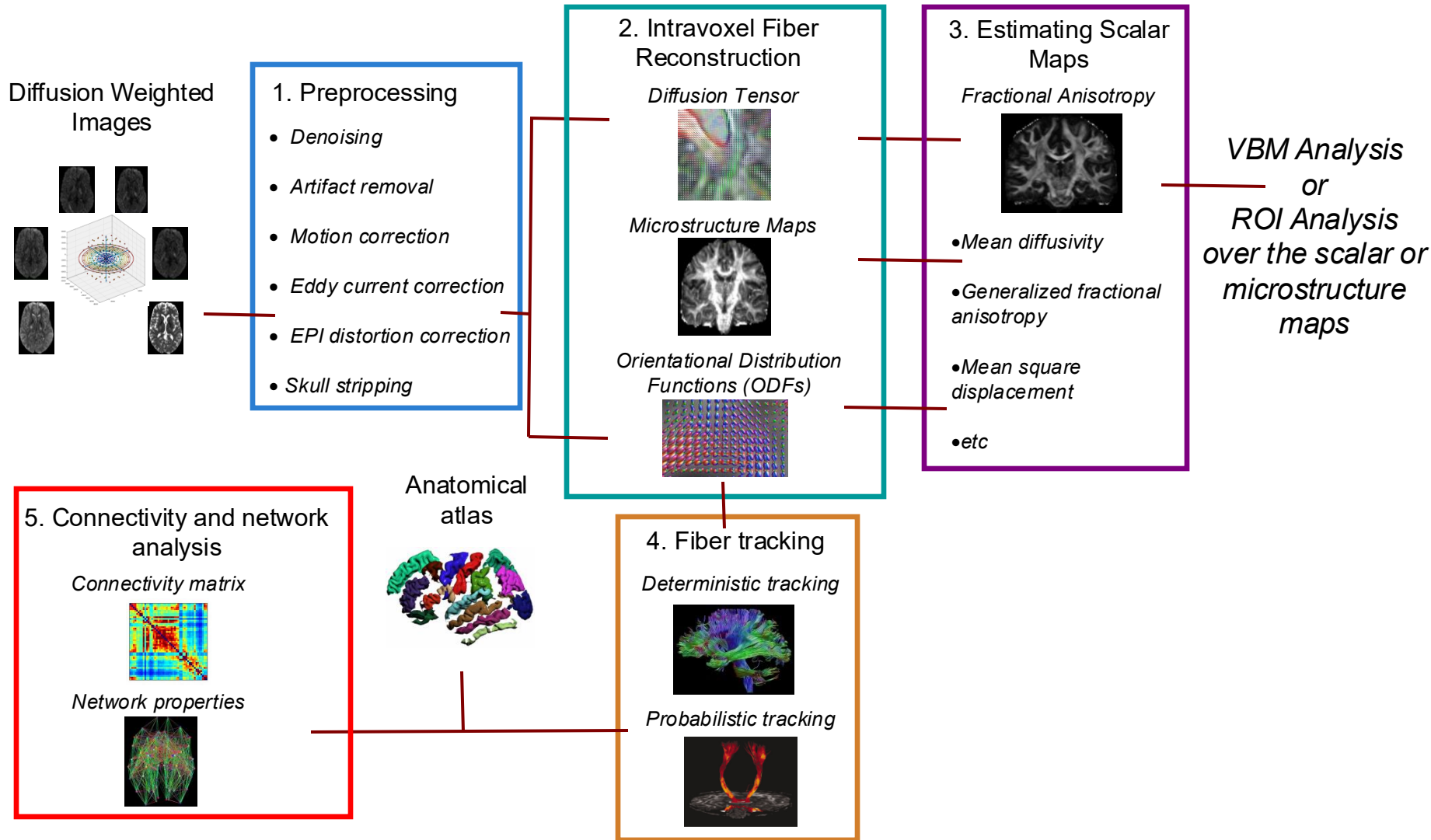
DTI: Diffusion Tensor Imaging

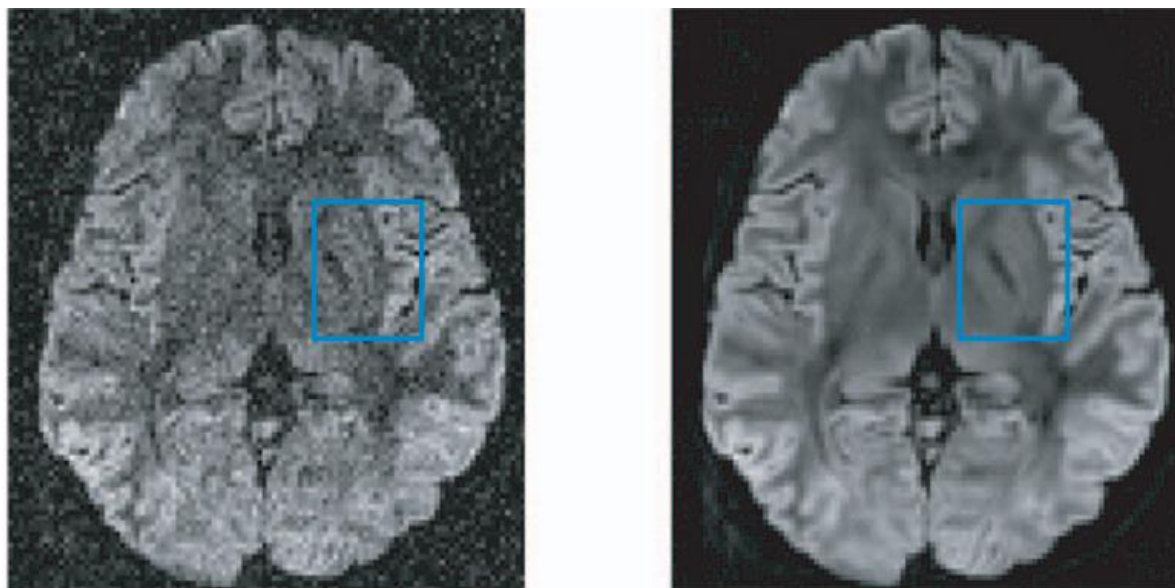
DSI: Diffusion Spectrum Imaging

HARDI: High Angular Resolution Diffusion Imaging

FSL: FMRIB Software Library ([FSL Web page](http://fsl.fmrib.ox.ac.uk/fsl/))

Global pipeline





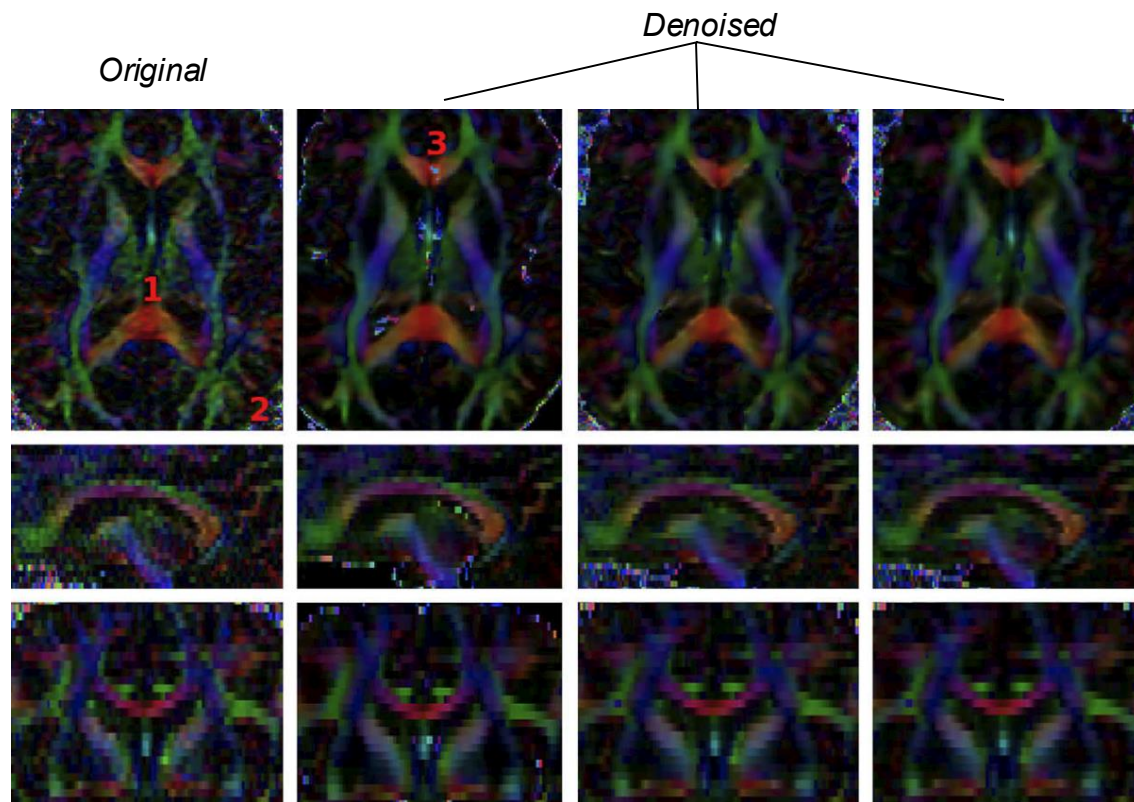
Zhang et al, 2017

Preprocessing

Preprocessing

■ Denoising

Mrtrix3 (dwidenoise, C++) Dipy (dipy.denoise.localpca, Python) GL-HOSVD (glhosvd, Matlab) 3D Slicer (C++)



Tristán-Vega et al, 2009

Preprocessing

■ Movement

- With increased scan acquisition time, subjects in the scanner may move more.
- Subject movement due to cardiac pulse and breathing

■ Eddy Currents

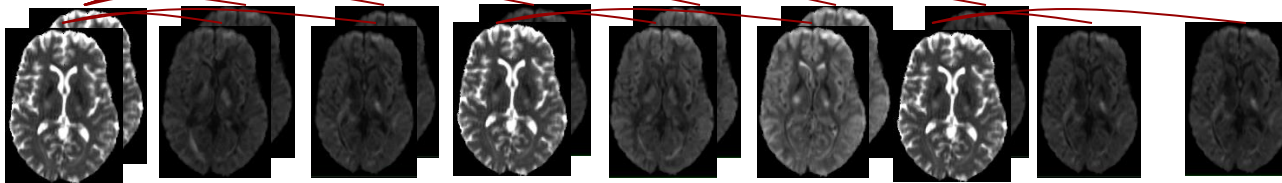
- Quick changes of the magnetic gradient produces induced currents in conductor structures of the scanner. Those induced currents produce induced magnetic fields (Faraday's Law)
 - Geometrical distortions

Affine registration (Translation + Rotation + Scaling + Shearing: 12 dof)

1. ~~Registration to the same volume of reference~~

a) ~~Registration to the closest B0~~

b) ~~Registration of all B0s to the volume of reference~~



Same tools than motion correction step

ANTs

(*antsRegistration*, C++)

SPM

(*spm_coreg*, Matlab)

FSL

(*flirt*, C++)

2. Apply the transformations to the gradient table

2. Apply the transformations to the gradient table

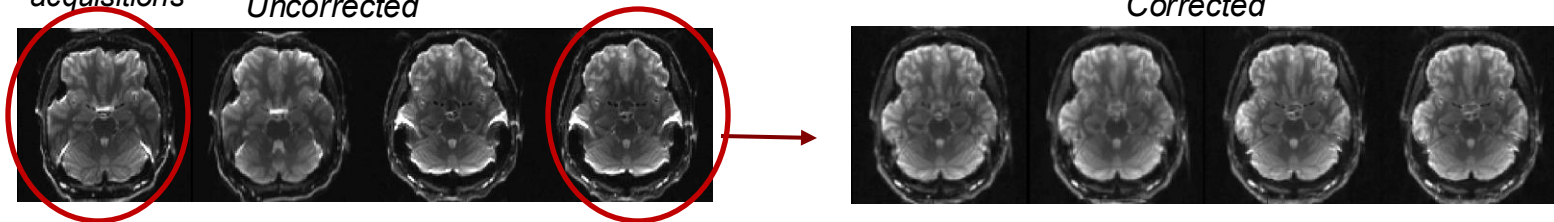
Susceptibility-induced distortion corrections

- Distortions because of magnetic susceptibility changes
 - Magnetic susceptibility changes at the interface between different tissues produce geometrical distortions

Susceptibility-Induced Distortion correction

- Using BlipUp and BlipDown acquisitions

TOPUP (eddy, C++)

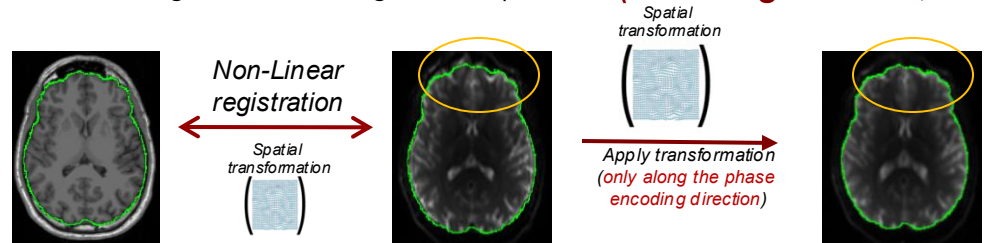


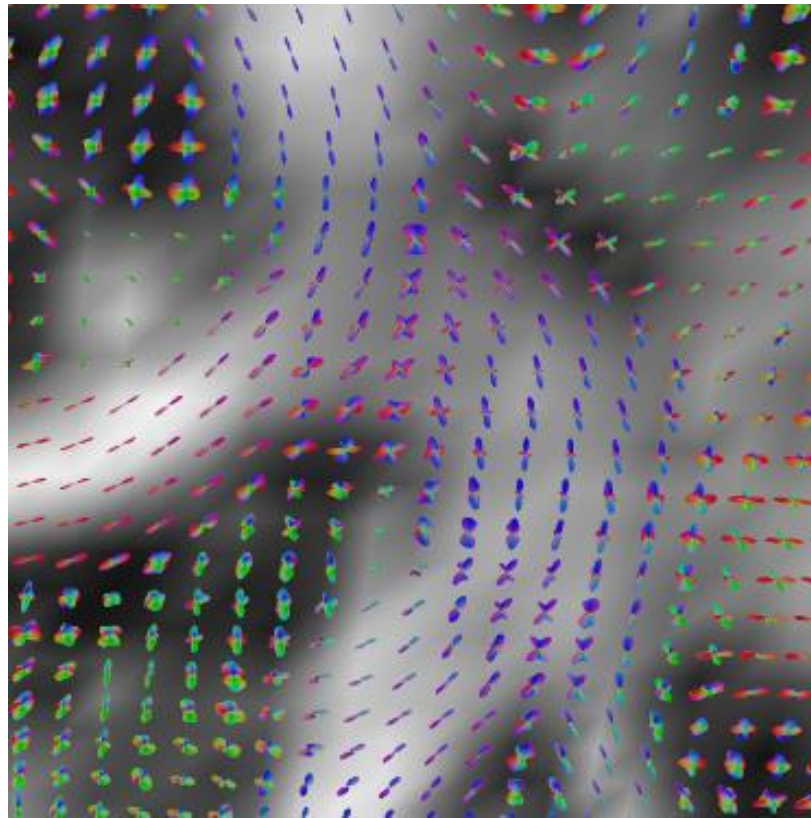
- Using Field Maps

FieldMap Toolbox
(FieldMap.m, Matlab)

FUGUE
(fugue, C++)

- Using Non-Linear registration (ANTs (antsRegistration, C++))





Intravoxel Fiber Reconstruction

Apparent Diffusion Coefficient (ADC)

$$S(\mathbf{g}) = S_0 e^{-bADC(\mathbf{g})} \Rightarrow ADC(\mathbf{g}) = -\frac{1}{b} \ln \frac{S(\mathbf{g})}{S_0}$$

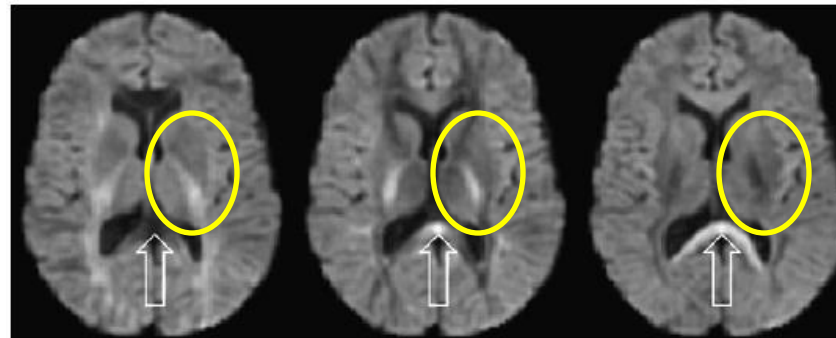
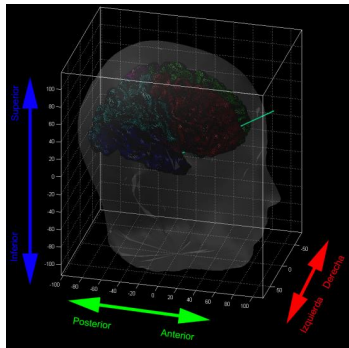
The apparent diffusion coefficient depends on the diffusion sensitizing gradient direction

Diffusion Weighted Images obtained for different diffusion sensitizing gradients

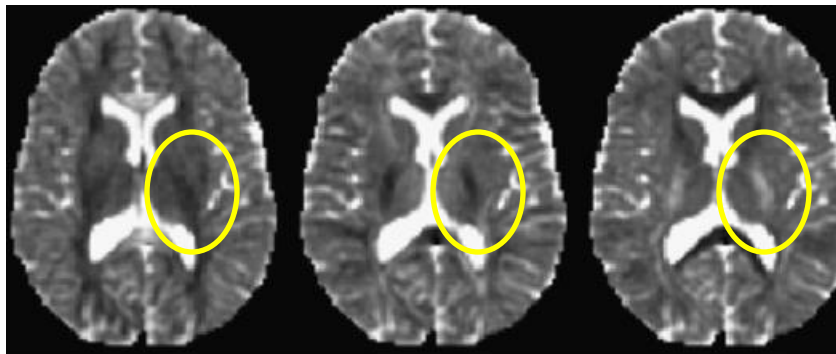
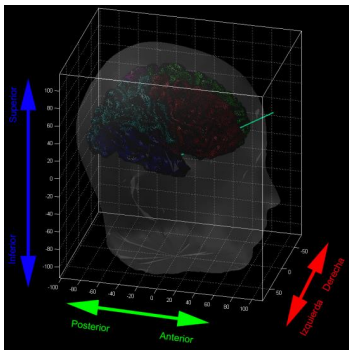
Left – Right.

Ant. - Post.

Inf. - Sup.

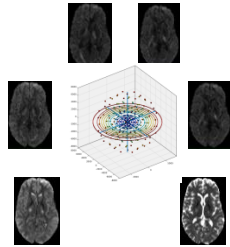


Apparent Diffusion Coefficients



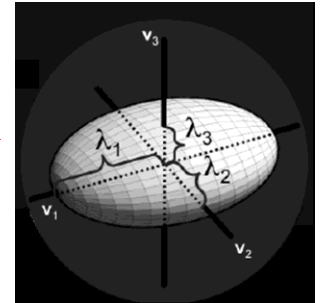
Intravoxel fiber reconstruction (Diffusion Tensor Imaging)

Diffusion Weighted Images

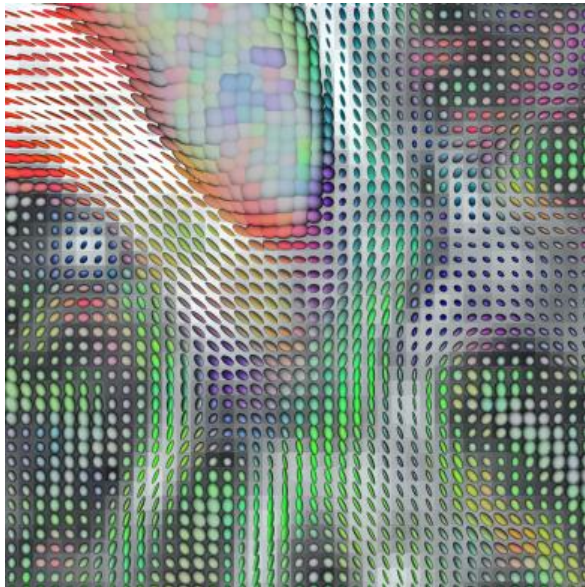


$$\mathbf{D} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yx} & D_{yy} & D_{yz} \\ D_{zx} & D_{zy} & D_{zz} \end{bmatrix} = \begin{bmatrix} V_{1x} & V_{2x} & V_{3x} \\ V_{1y} & V_{2y} & V_{3y} \\ V_{1z} & V_{2z} & V_{3z} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} V_{1x} & V_{2x} & V_{3x} \\ V_{1y} & V_{2y} & V_{3y} \\ V_{1z} & V_{2z} & V_{3z} \end{bmatrix}^T$$

Eigen decomposition



Diffusion Tensors



Some freely available tools

- *Least square estimation*
- *Weighted least square estimation*
- *Symmetric Positive-Definite Constraints*

FSL
(*dtifit*, C++)

Mrtrix3
(*dwi2tensor*, C++)

Diffusion Toolkit
(*DTI-TK*, C++)

DSI-Studio
(C++)

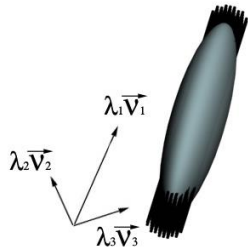
Camino
(*modelfit*, C++)

Dipy
(*dipy.reconst.dti*, Python)

Diffusion Tensor Estimation with Symmetric
Positive-Definite Constraints (*Matlab*)

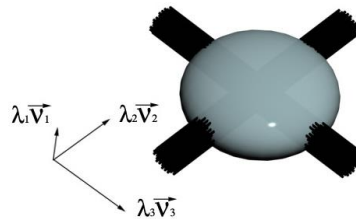
Diffusion Tensor Imaging (DTI)

Uniaxial model



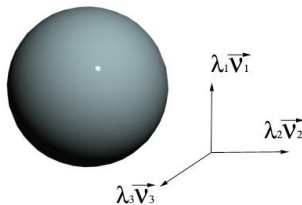
Planar model

$$\lambda_2 \sim \lambda_3 \gg \lambda_1$$



Spherical model

$$\lambda_1 \sim \lambda_2 \sim \lambda_3$$



Fractional Anisotropy and Mean Diffusivity

$$FA = \sqrt{\frac{3}{2}} \sqrt{\frac{(\lambda_1 - MD)^2 + (\lambda_2 - MD)^2 + (\lambda_3 - MD)^2}{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}},$$

$$MD = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$$

Fractional Anisotropy



Main eigenvector



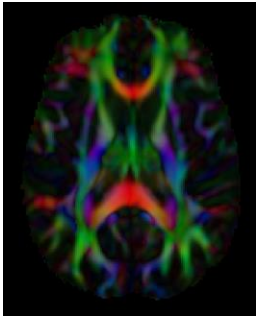
Color-coded Fractional Anisotropy



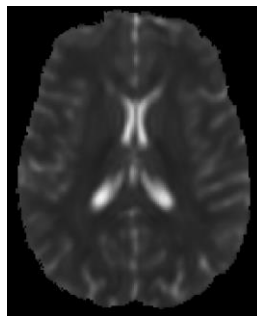
Scalar maps from diffusion tensor imaging

Scalar maps that can be extracted from diffusion tensor imaging

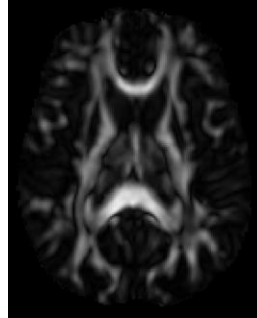
Fractional anisotropy
(FA)



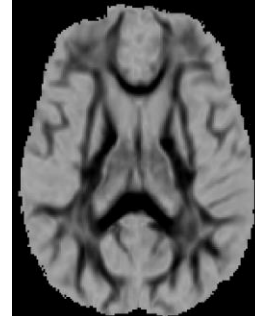
Mean diffusivity
(MD)



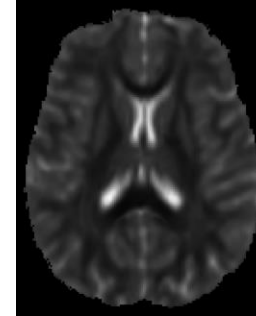
Linear coefficient
(LC)



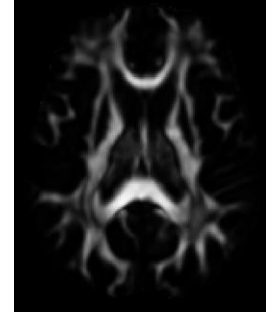
Spherical coefficient
(SC)



Radial diffusivity
Radial (RD)



Volume fraction
(VF)



Some freely available tools

FSL

(*dtifit*, C++)

Mrtrix3

(*tensor2metric*, C++)

Dipy

(*dipy.reconst.dti*, Python)

Camino

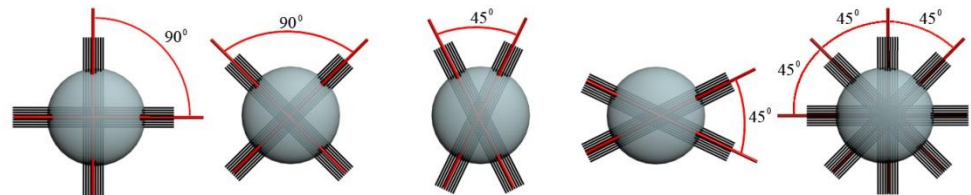
(*modelfit*, C++)

Advantages

- Short acquisition time
- Low computation times
- Easy interpretation

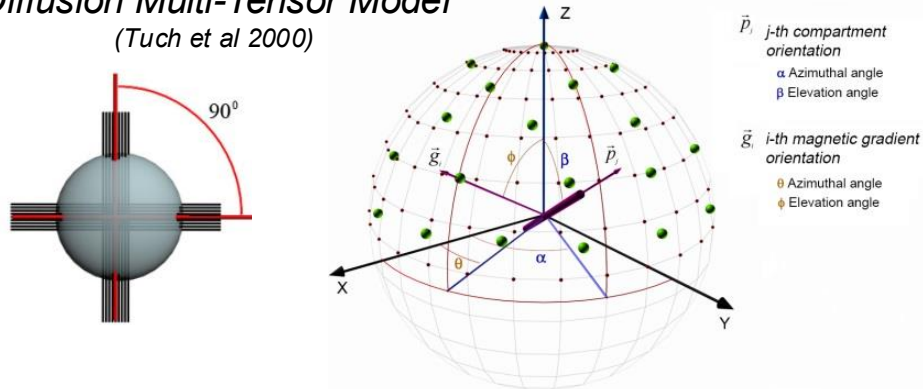
Drawbacks

- Not reliable in regions with fiber crossings, kissing, bending, etc

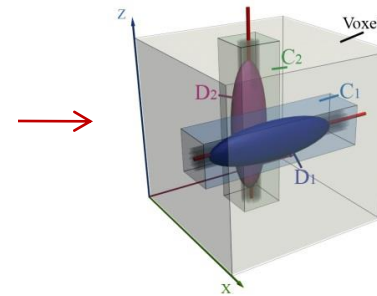


Intravoxel fiber orientation

Diffusion Multi-Tensor Model (Tuch et al 2000)



Multiple compartments inside the voxel



$$S_1(b, \vec{u}) = S_0 \exp[-b \vec{u} \cdot \mathbf{D}_1 \cdot \vec{u}^T]$$

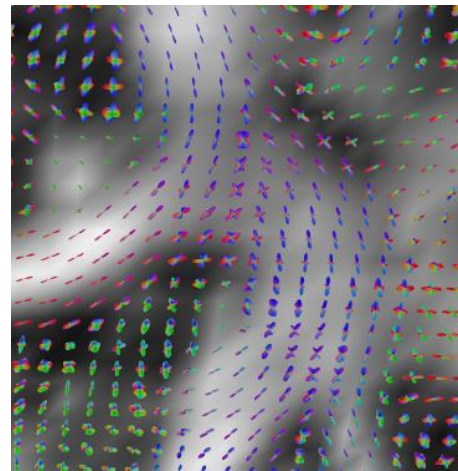
$$S_2(b, \vec{u}) = S_0 \exp[-b \vec{u} \cdot \mathbf{D}_2 \cdot \vec{u}^T]$$

$$S_t = \underbrace{f_1 S_1 + f_2 S_2}_{\text{Volume fractions}}$$

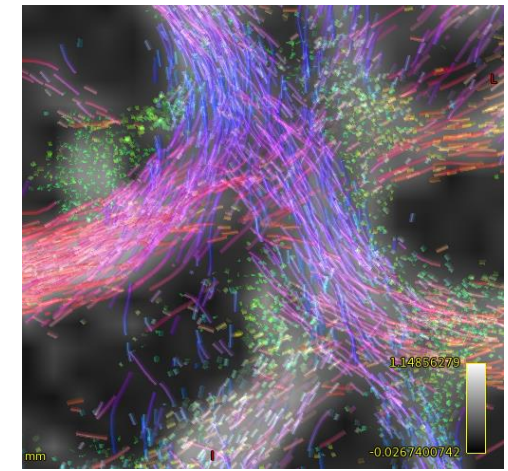
Diffusion Tensors



Orientational Distribution Functions (Tuch et al 2004)



Fiber tracking



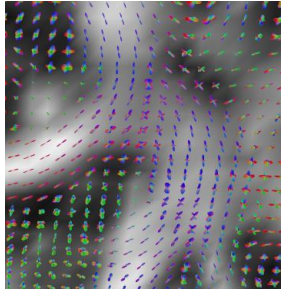
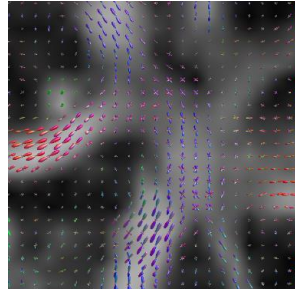
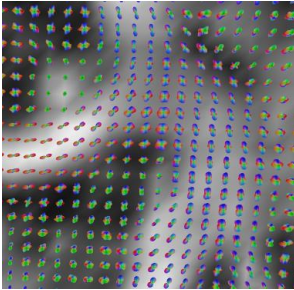
Intravoxel fiber reconstruction (ODFs)

QBall

CSD

RUMBA

HARDI data

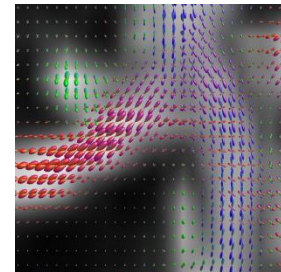
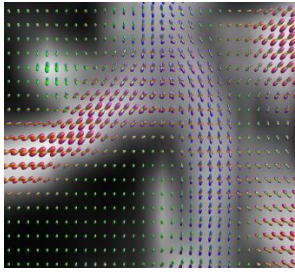
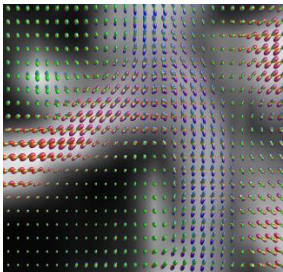


DSI

SHORE

DSI with deconvolution

DSI data



Available tools for DSI data

- **DSI reconstruction (DSI)**
 - [DTK](#) ([odf_recon](#), [C++](#))
 - [pHARDI](#) ([phardi](#), [C++](#))
 - [Dipy](#) ([DiffusionSpectrumModel](#), [Python](#))
- **DSI with deconvolution (DSIdeconv)**
 - [Dipy](#) ([DiffusionSpectrumDeconvModel](#), [Python](#))
- **SHORE (DSIdeconv)**
 - [Dipy](#) ([ShoreModel](#), [Python](#))

Available tools for HARDI data

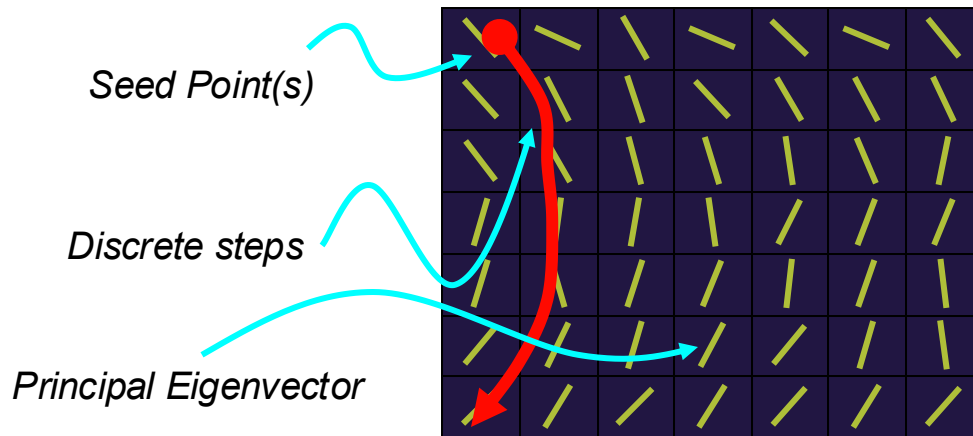
- **Qball reconstruction (QBall)**
 - [DTK](#) ([odf_recon](#), [C++](#))
 - [FSL](#) ([qboot](#), [C++](#))
 - [pHARDI](#) ([phardi](#), [C++](#))
 - [DSI-Studio](#) ([qbi](#), [Matlab](#) and [C++](#))
- **Qball Constrained Solid Angle (QBall_CSA)**
 - [FSL](#) ([qboot](#), [C++](#))
 - [pHARDI](#) ([phardi](#), [C++](#))
 - [Dipy](#) ([CsaOdfModel](#), [Python](#))
- **Generalized Q-sampling Imaging (GQI)**
 - [pHARDI](#) ([phardi](#), [C++](#))
 - [DSI-Studio](#) ([gqi_I](#), [Matlab](#) and [C++](#))
- **Diffusion Orientation Transform Revisited**
 - [pHARDI](#) ([phardi](#), [C++](#))
- **RUMBA**
 - [pHARDI](#) ([phardi](#), [C++](#))
- **Constrained Spherical Deconvolution (CSD)**
 - [Mrtrix3](#) ([dwi2response](#) and [dwi2fod](#), [C++](#))
 - [Dipy](#) ([csdeconv](#), [Python](#))
- **Mean Apparent Propagator MRI (MAP-MRI)**
 - [Dipy](#) ([mapmri](#), [Python](#))
- **Bedpostx**
 - [FSL](#) ([bedpostx](#), [C++](#))



Fiber Tracking

Fiber Tracking

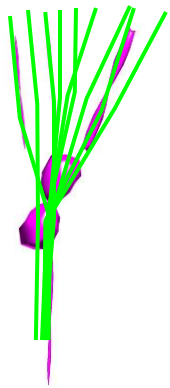
■ Deterministic Tracking



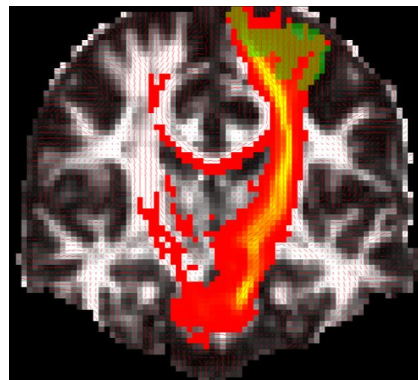
Streamlines



■ Probabilistic Tracking



Connection probability



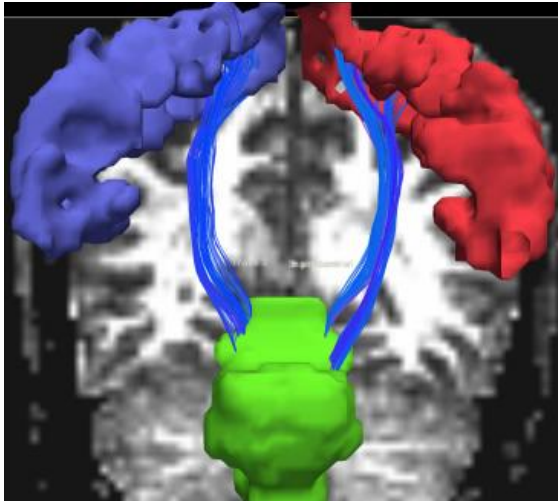
■ Main drawback

Bias with the length of the bundles

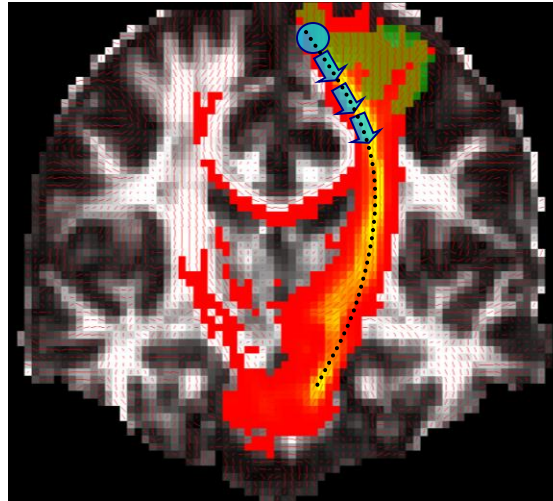
Fiber tracking

Probabilistic tractography

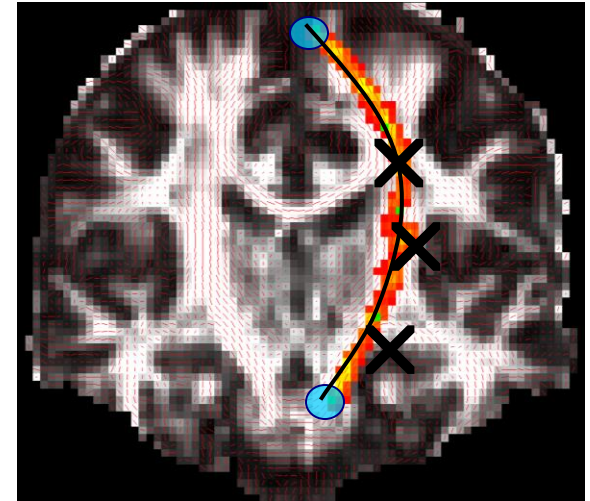
Deterministic tractography



Local tractography



Global tractography



Available tools for fiber tracking

■ Deterministic tracking

DTK (*odf_tracker*, C++)

Mrtrix3 (*tckgen FACT*, C++)

Mrtrix3 (*tckgen SD_STREAM*, C++)

Mrtrix3 (*tckgen Tensor_Det*, C++)

DSI-Studio (Matlab and C++)

Dipy (*dipy.tracking.eudx*, Python)

■ Probabilistic tracking with crossing fibres (Probtrackx)

FSL (*probtrackx*, C++)

■ Probabilistic Index of Connectivity (Pico)

CAMINO (*track*, C++)

■ Probabilistic tracking using Mrtrix

Mrtrix3 (*tckgen iFOD1*, C++)

Mrtrix3 (*tckgen iFOD2*, C++)

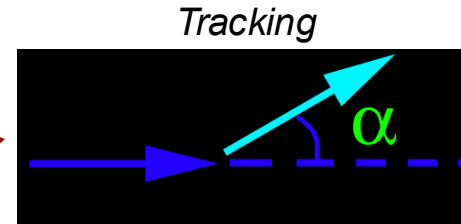
Mrtrix3 (*tckgen Tensor_Prob*, C++)

FACT: Fiber Assigned by Continuous Tracking

Important parameters for fiber tracking

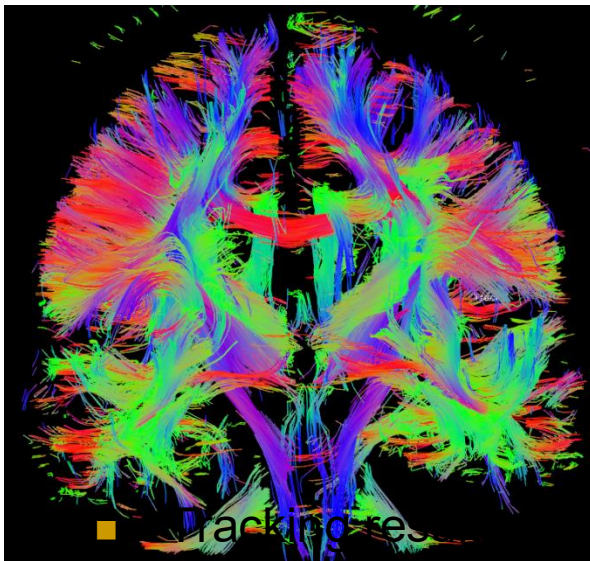
■ Fiber tracking parameters

- Fractional anisotropy threshold
- Fiber tracking step size
- Angular threshold

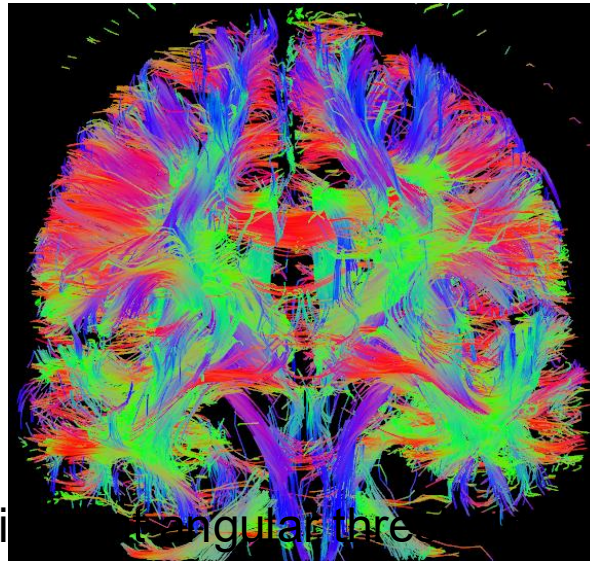


■ Tracking results for different angular thresholds

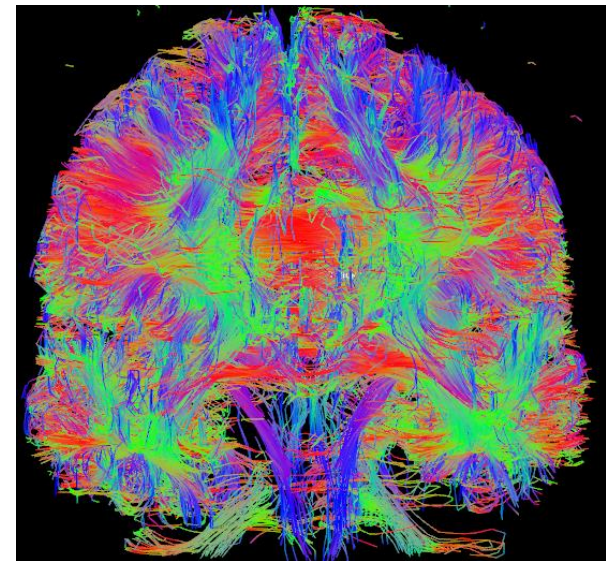
$\alpha = 45^\circ$



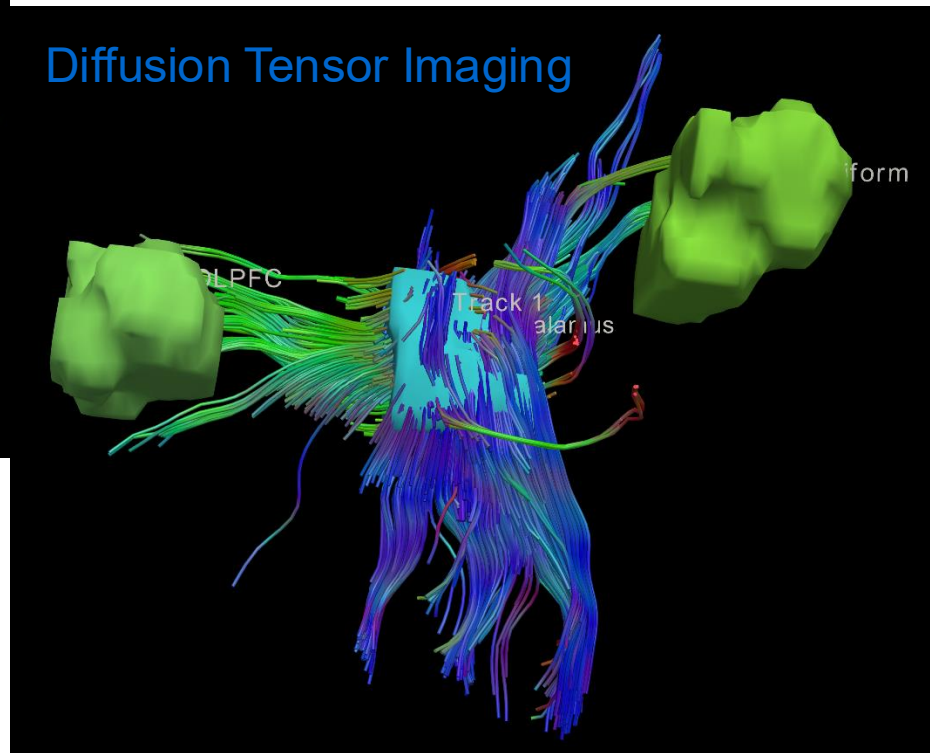
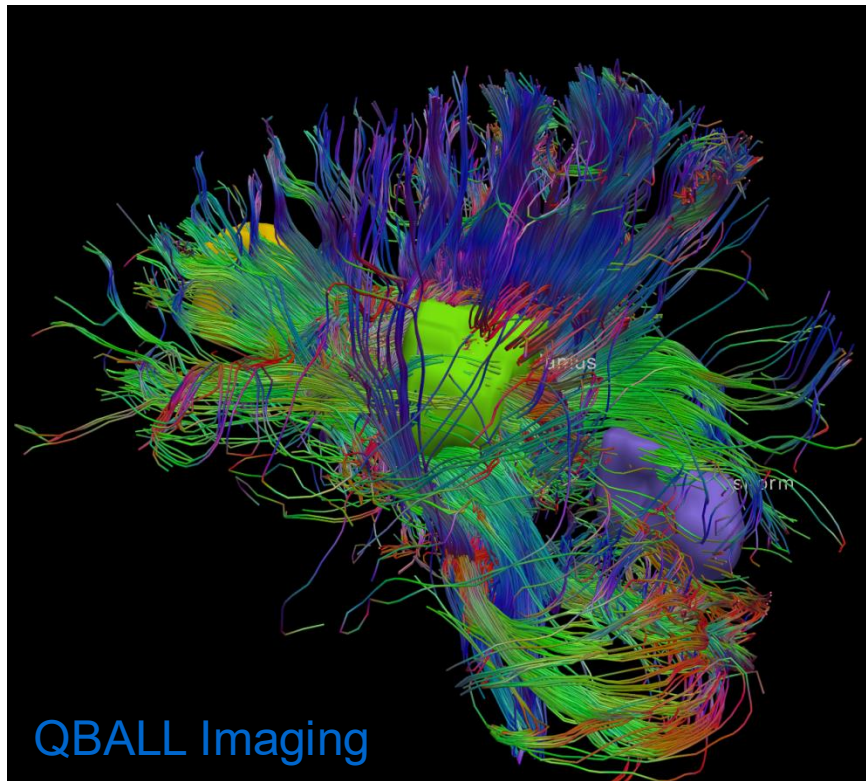
$\alpha = 65^\circ$



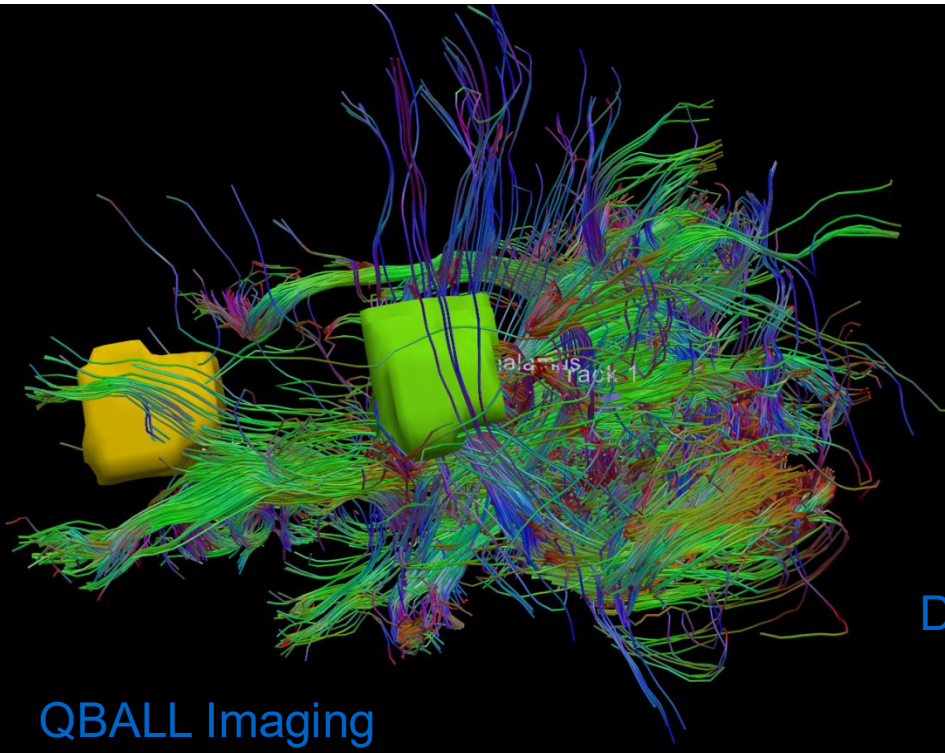
$\alpha = 85^\circ$



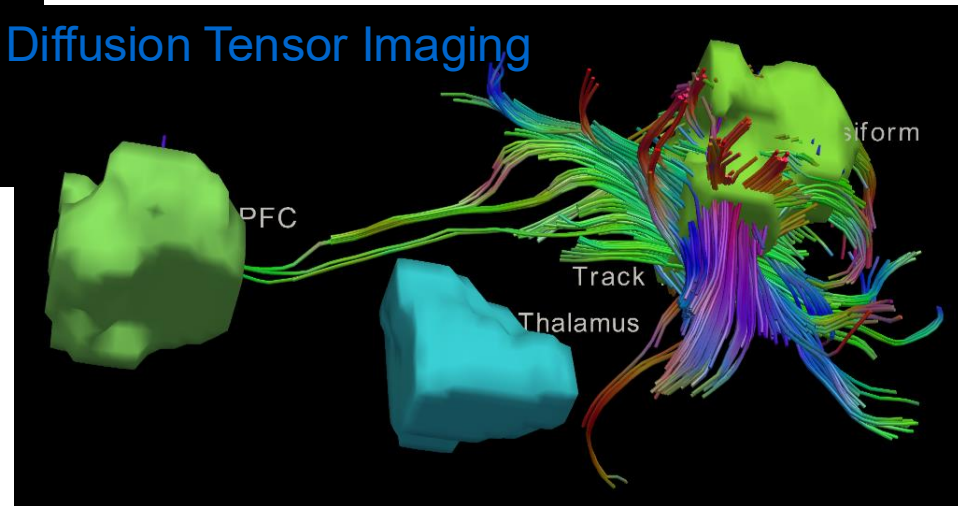
Thalamus vs Whole Brain



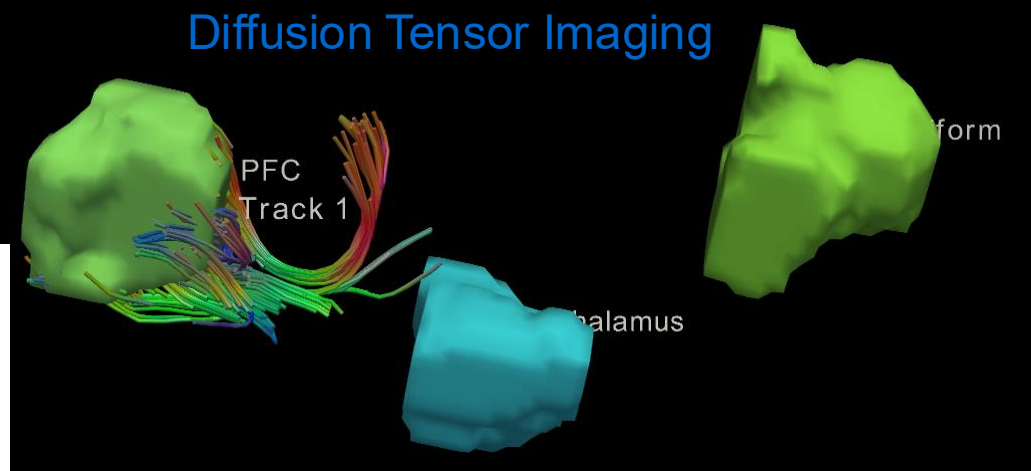
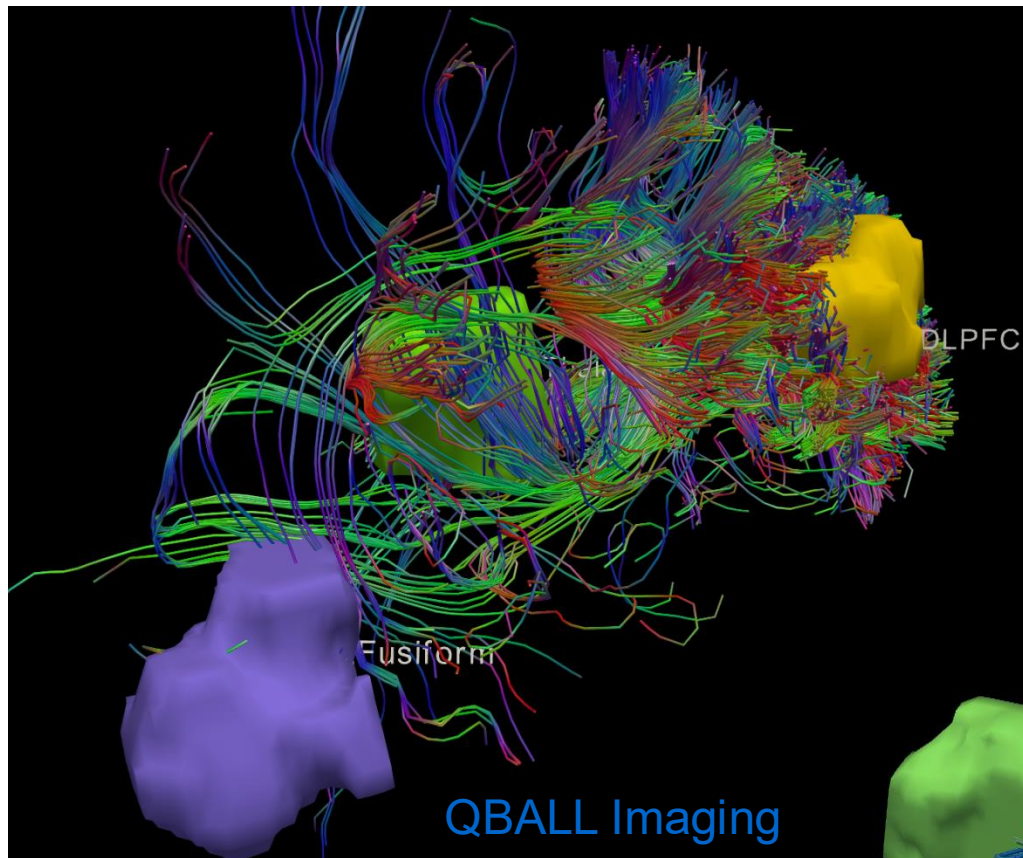
Fusiform Gyrus vs Whole Brain



Diffusion Tensor Imaging

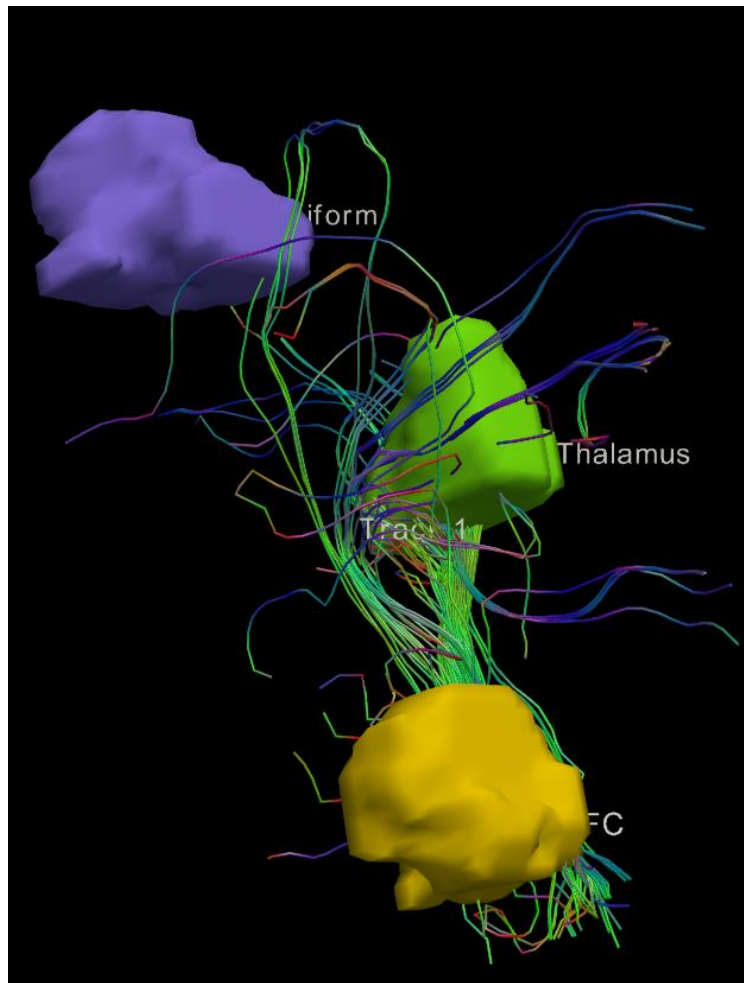


Dorso-Lateral Prefrontal Cortex vs Whole Brain

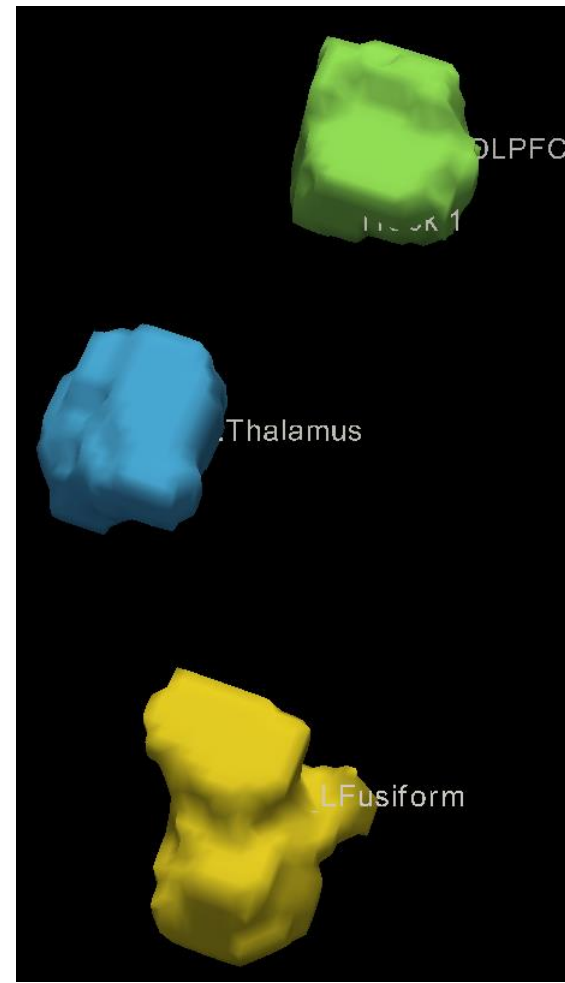


Dorso-Lateral Prefrontal Cortex vs Thalamus

QBALL Imaging

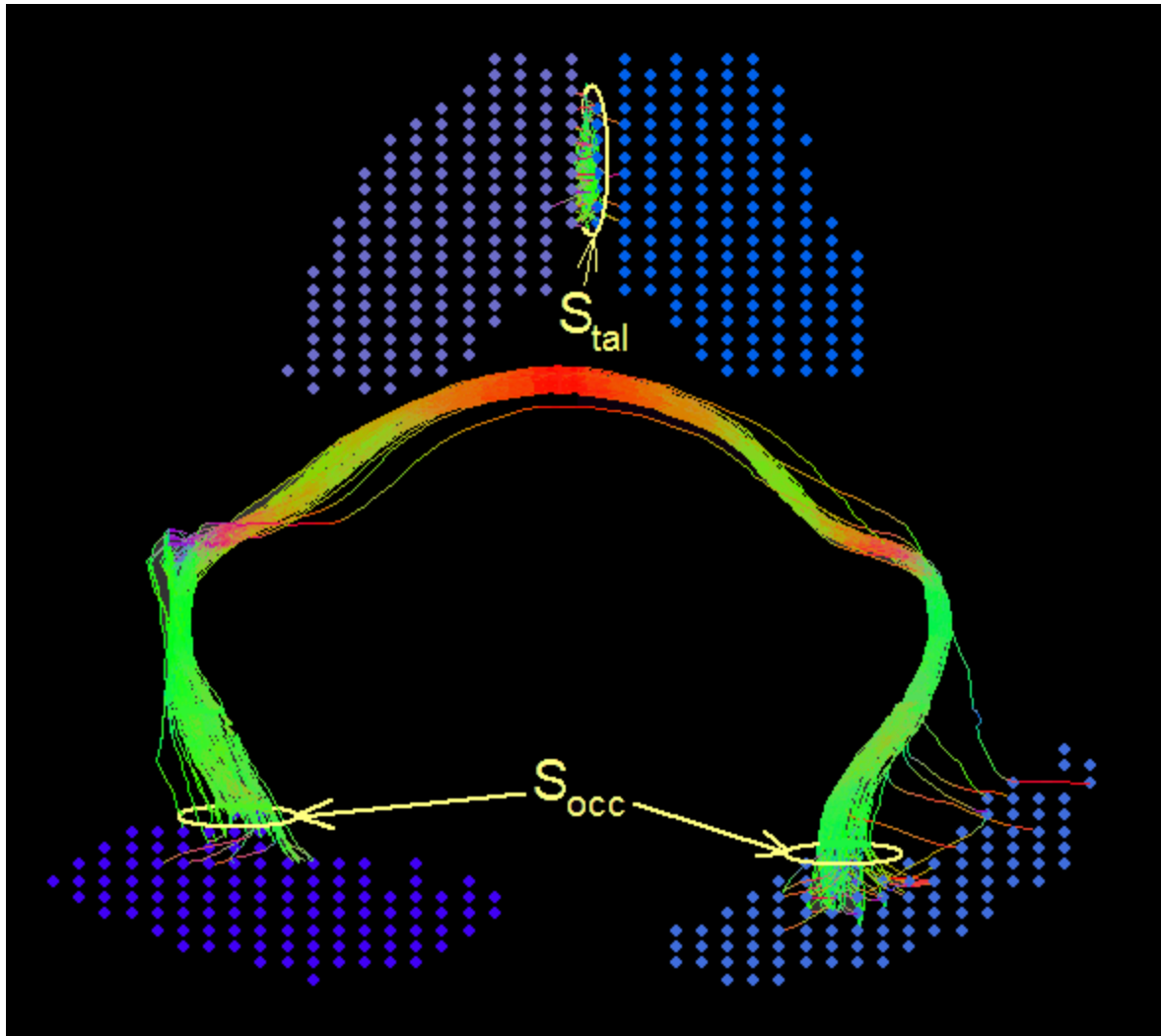


Diffusion Tensor Imaging



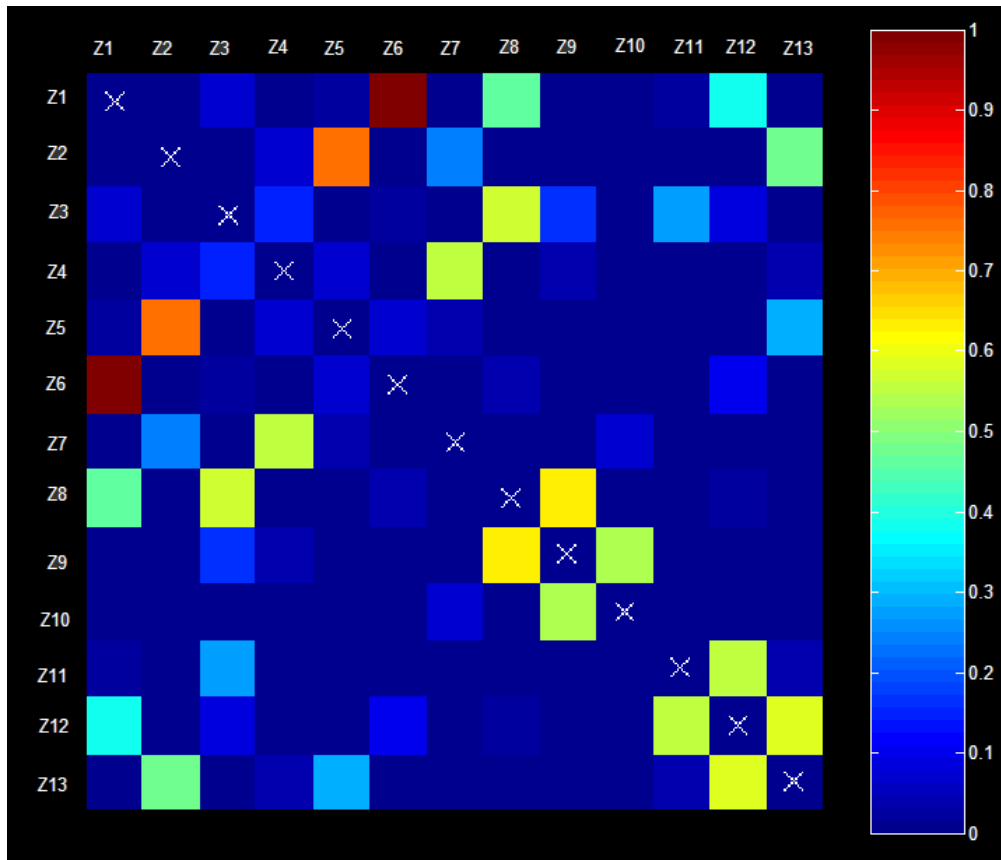
Structural Connectivity

Anatomical connectivity

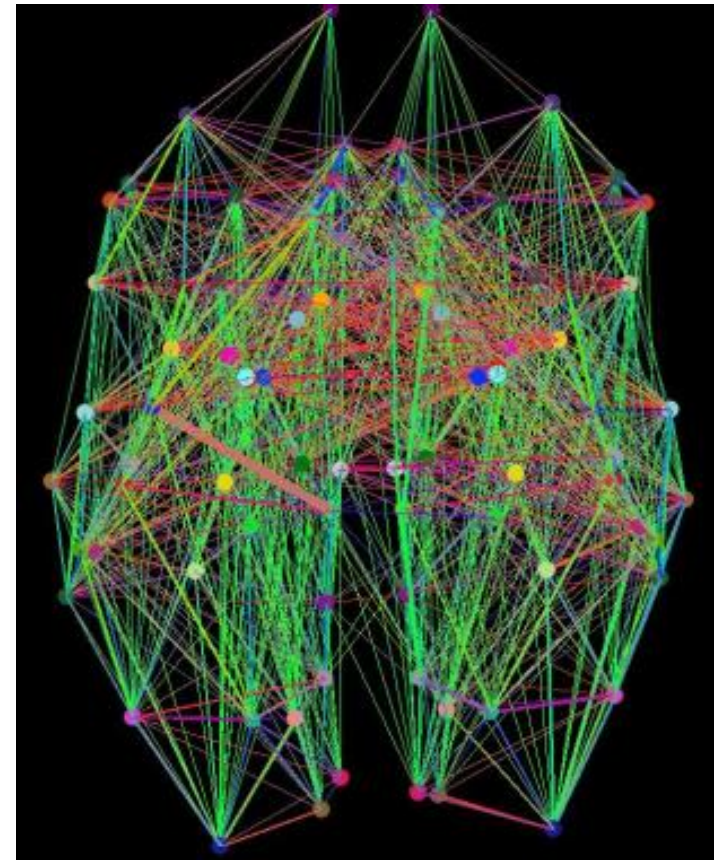


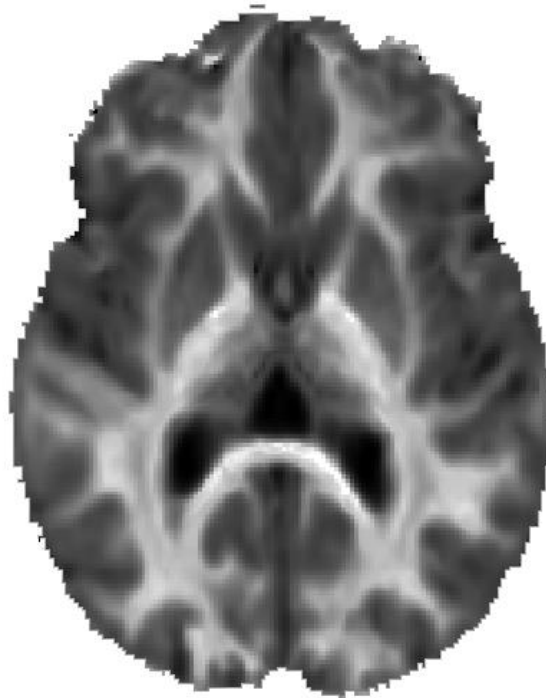
Connectivity Matrix

Connectivity matrix



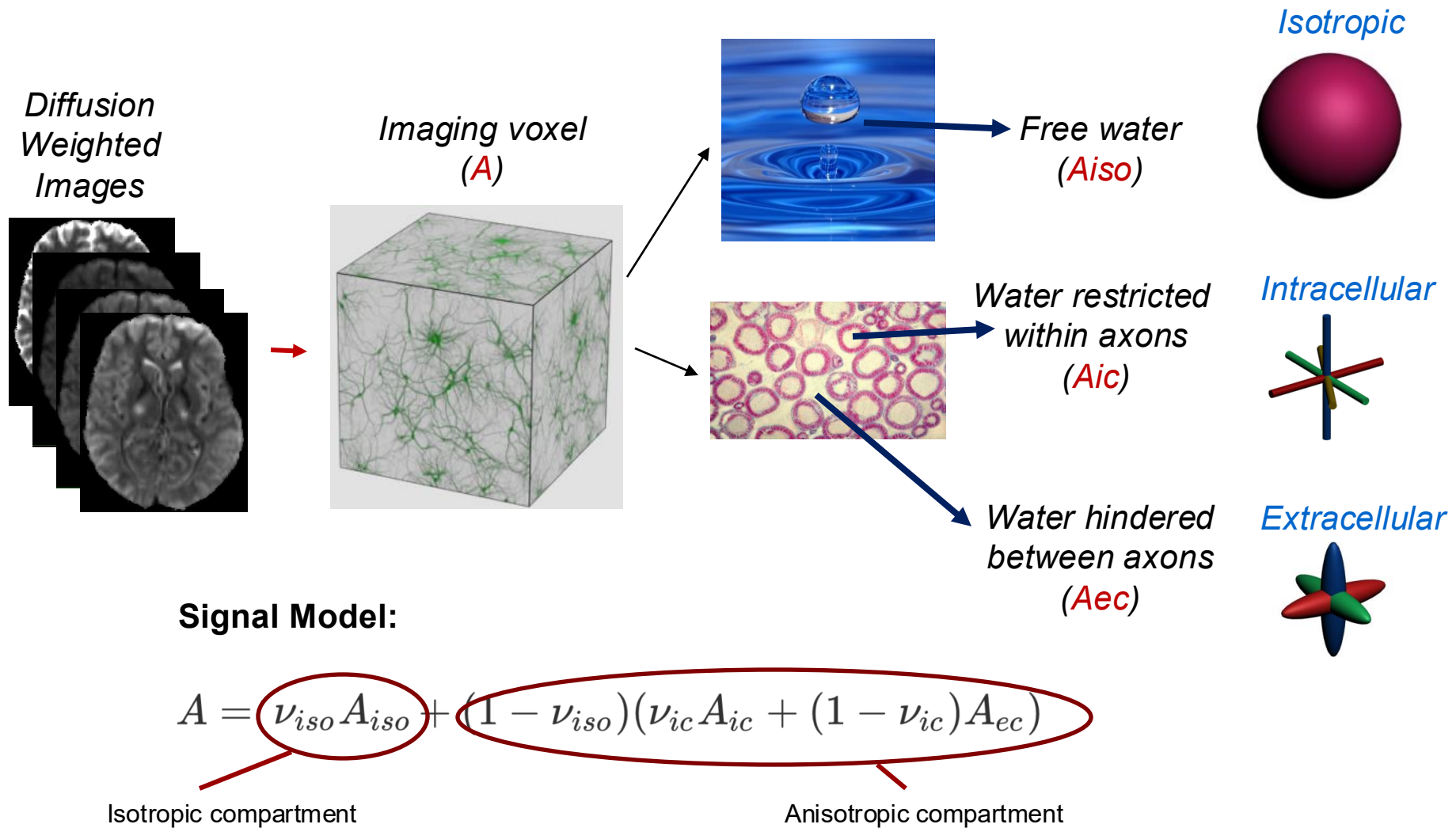
Graph representation





Microstructure

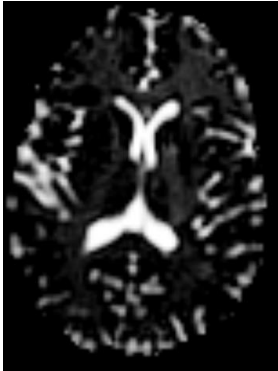
Neurite Orientation Dispersion and Density Imaging (NODDI) ¹



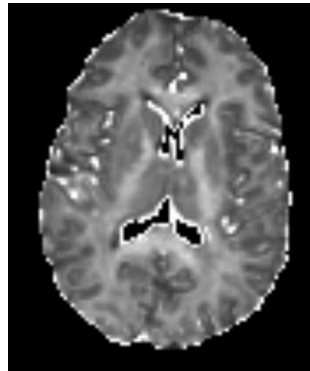
Microstructure maps

Neurite Orientation Dispersion and Density Imaging ([NODDI](#), [AMICO](#), [Python](#))

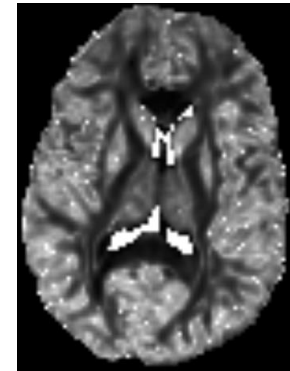
Isotropic volume fraction



Intracellular volume fraction

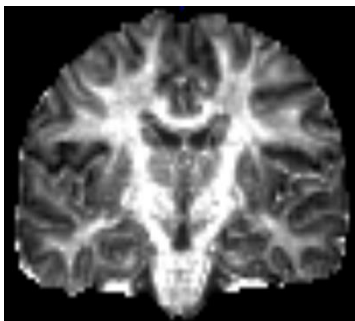


Orientation dispersion index

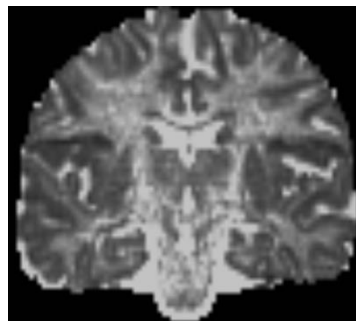


Spherical Mean Technique ([SMT](#), [smt](#), [C++](#))

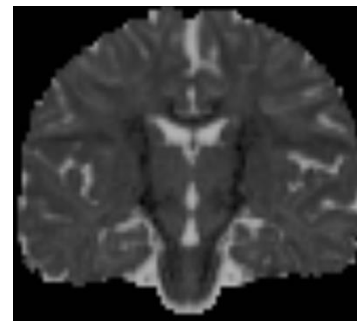
*Intra-neurite
volume fraction*



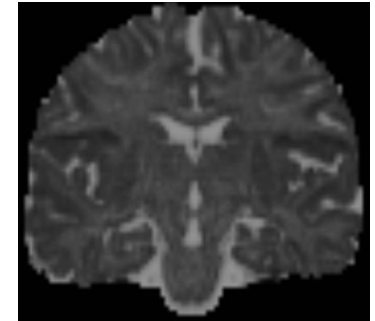
Intrinsic diffusivity

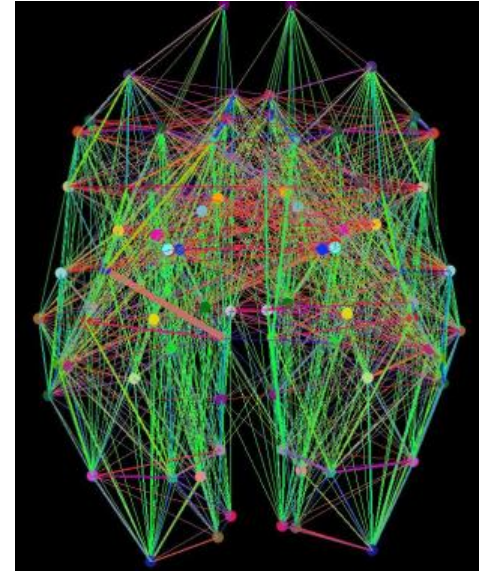
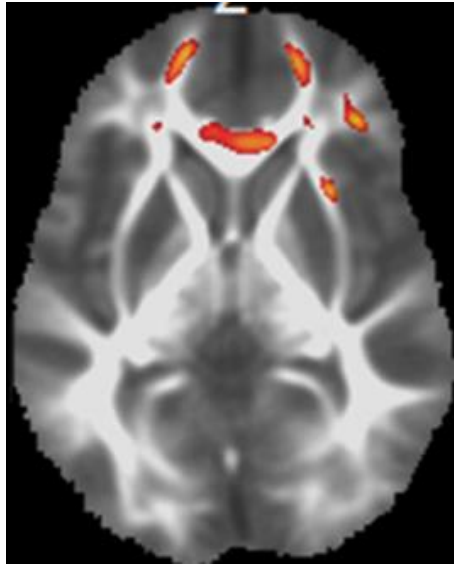
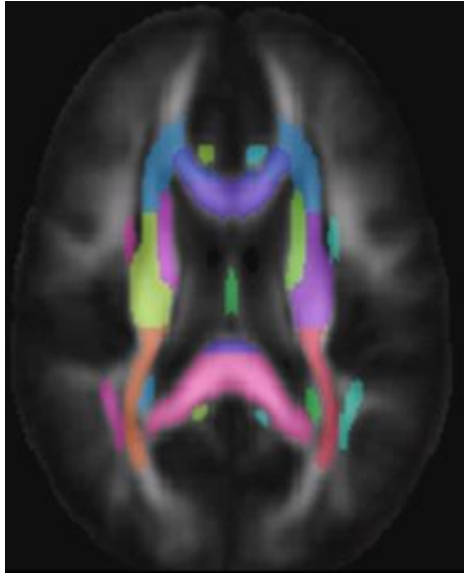


*Extra-neurite transverse
microscopic diffusivity*



*Extra-neurite microscopic
mean diffusivity*

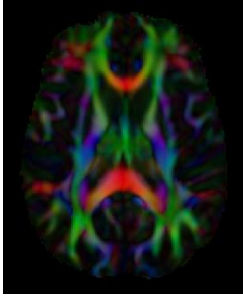




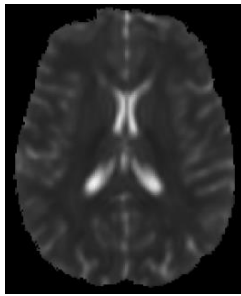
ROI analysis, Voxel based morphometry and Network analysis

ROI Analysis

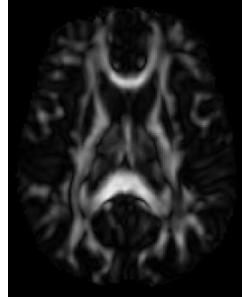
Fractional anisotropy



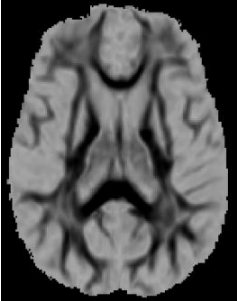
Mean diffusivity



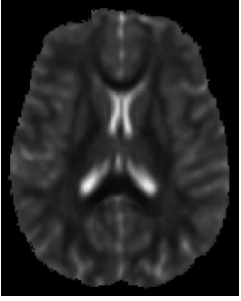
Linear coefficient



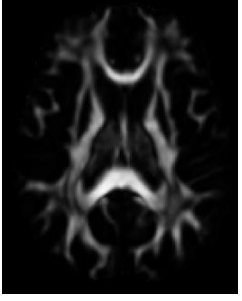
Spherical coefficient



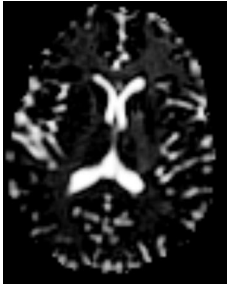
Radial diffusivity



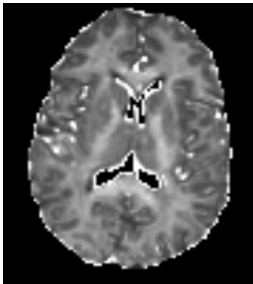
Volume fraction



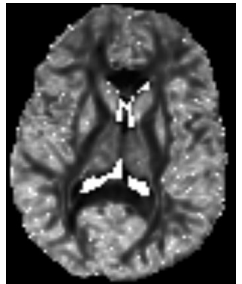
Isotropic volume fraction



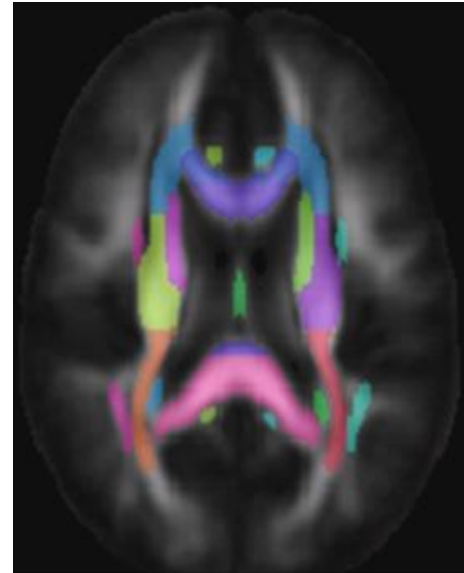
Intracellular volume fraction



Orientation dispersion index

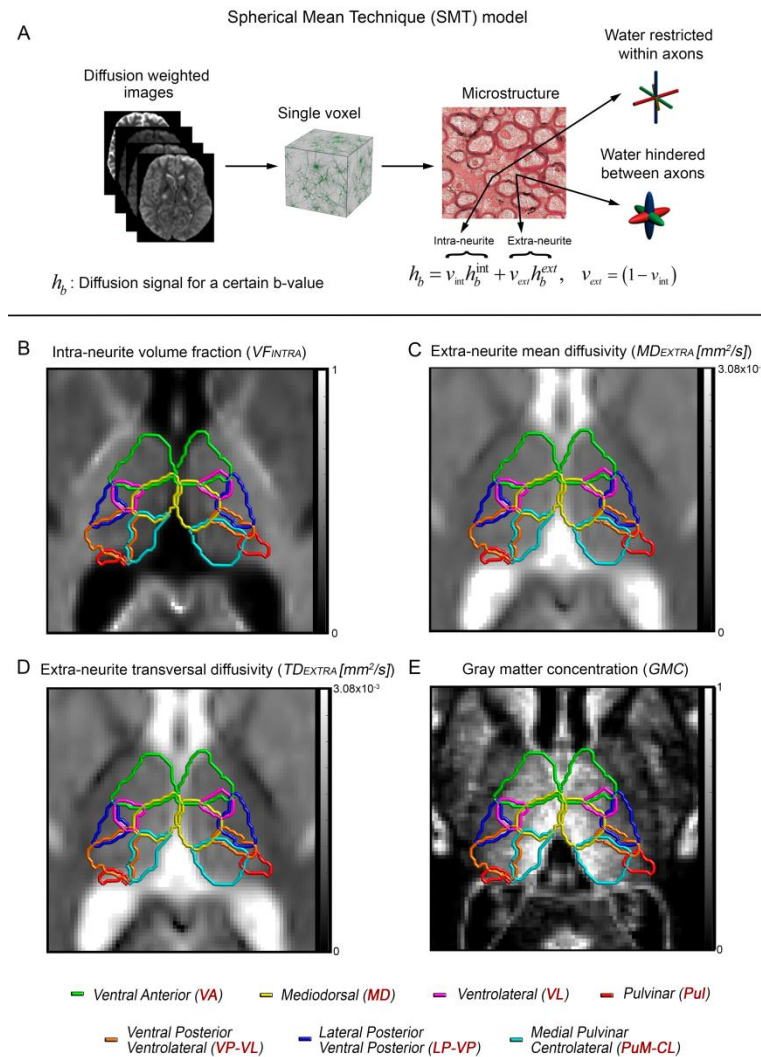


Region Of Interest



Values for each region and perform the statistics

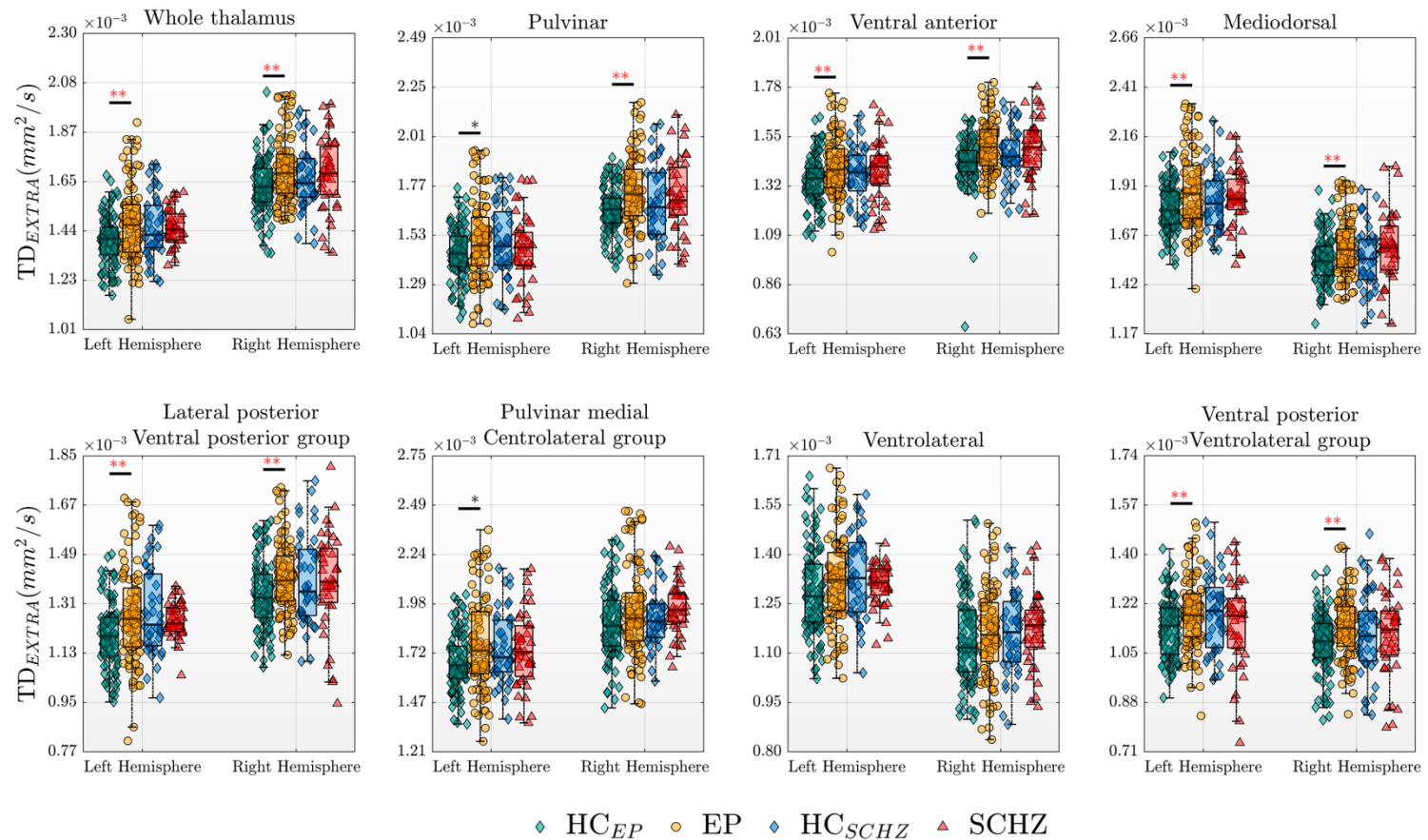
Assessing thalamic microstructure in psychosis



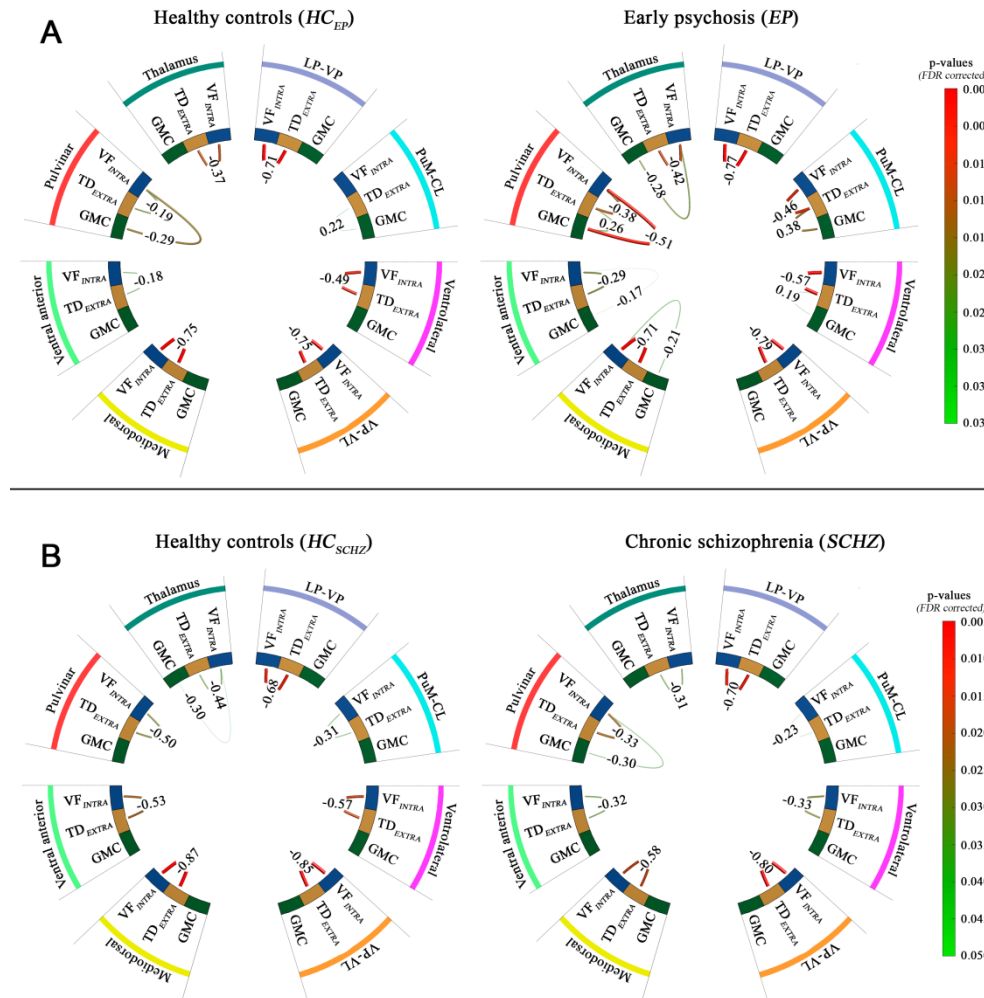
Assessing thalamic microstructure in psychosis

B

Group differences: Mean extra-neurite transversal diffusivity (TD_{EXTRA})

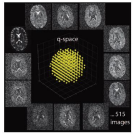


Assessing thalamic microstructure in psychosis



Voxel-based morphometry analysis

Diffusion Weighted Images



Movement correction

Eddy current correction

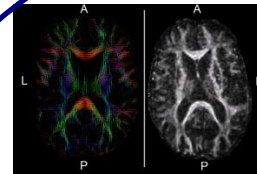
EPI distortion correction

Computing Scalar Maps (MD, FA, etc)

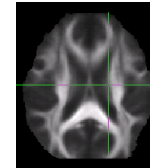
Spatial Normalization

Smoothing

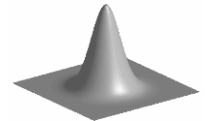
Individual pre-processing



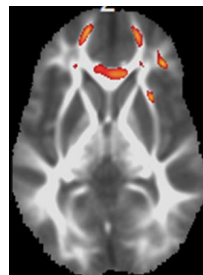
Stereotactic Space



Smoothing Kernel

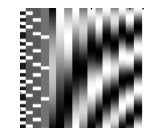


Statistical map



Statistical processing

General Linear Model



Design Matrix

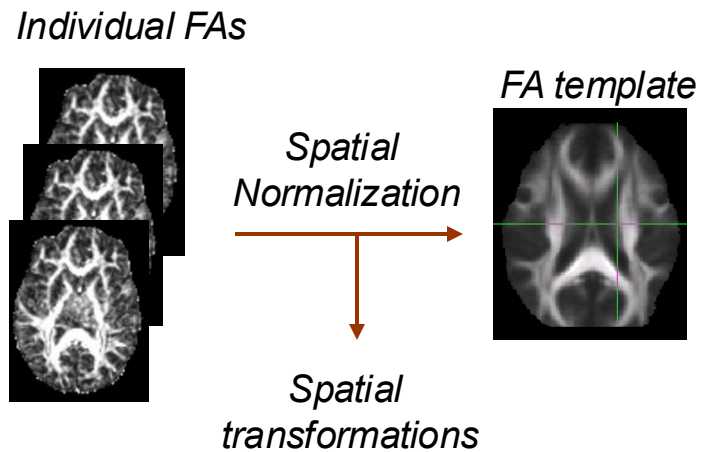
Group-wise processing

FA: Fractional Anisotropy

MD: Mean Diffusivity

EPI: Echo Planar Imaging

Spatial Normalization strategies

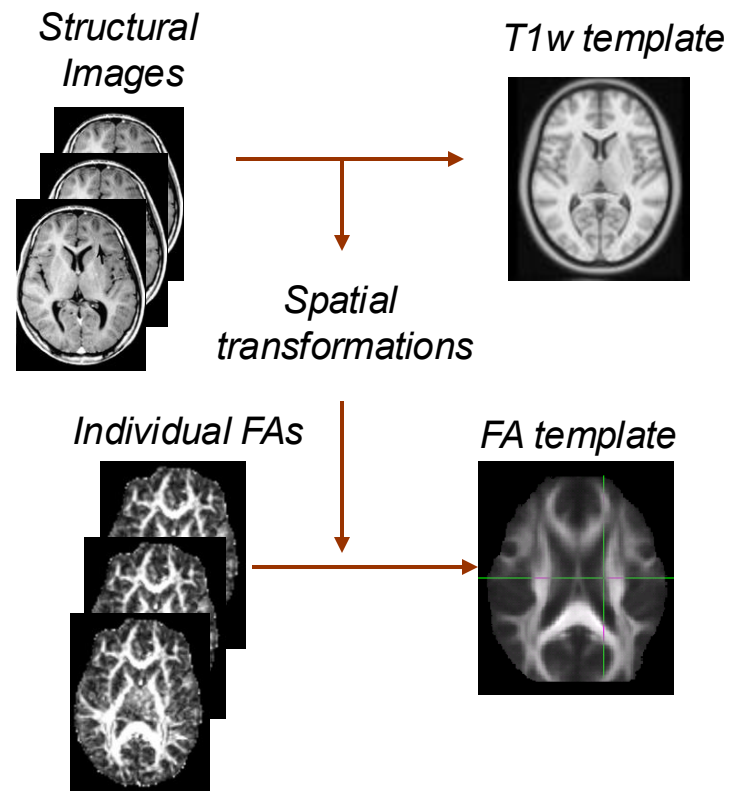


- **Advantages**

- Same modality

- **Drawbacks**

- Low spatial details in cortical regions
- Affected regions in FA can cause misregistration .
- Low spatial resolution



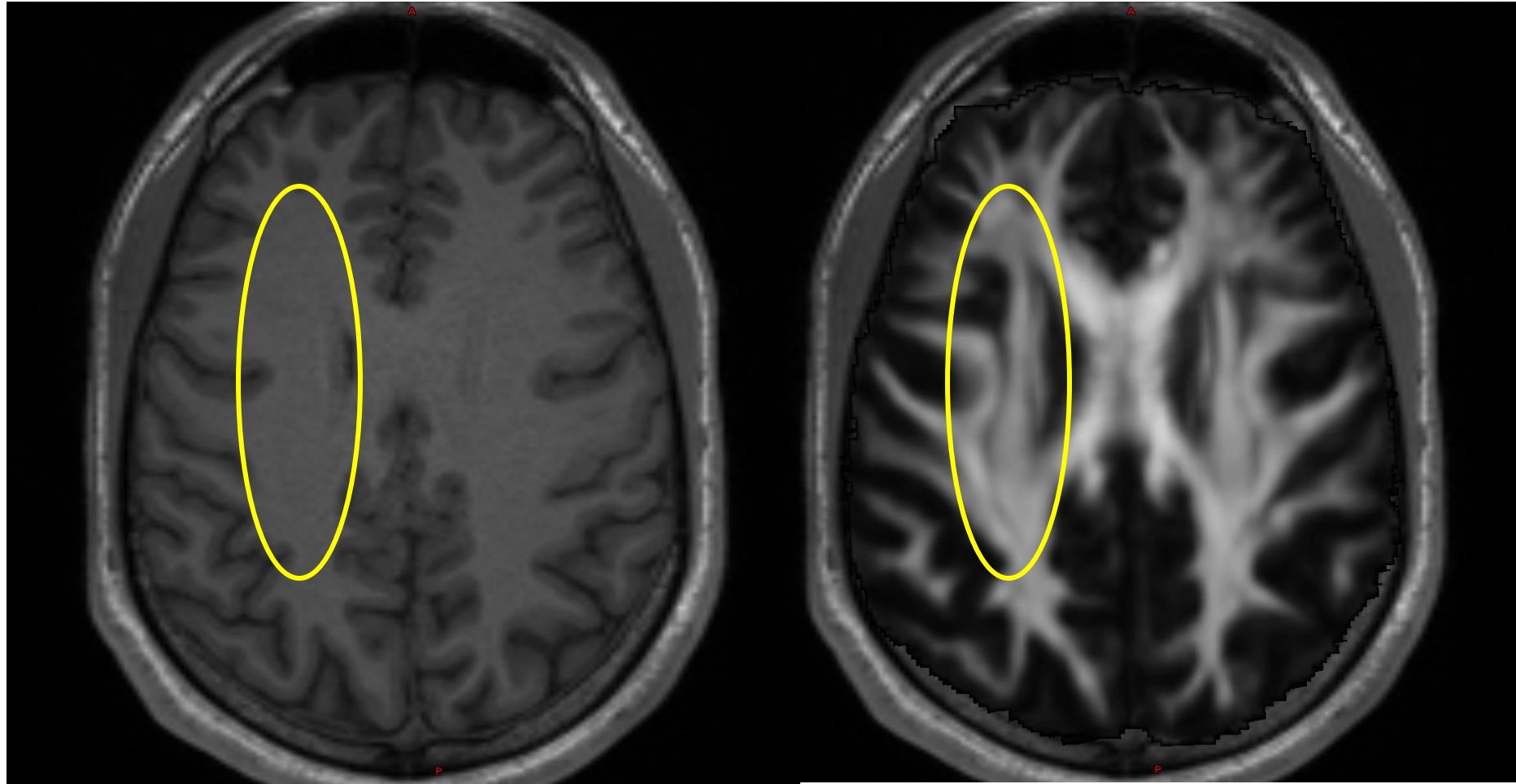
- **Advantages**

- High spatial resolution and structural details
- Robust (FA atrophy does not affect the alignment)

Spatial normalization strategies

T1w Image

FA Image



Meta-analysis about using VBM in DTI



Contents lists available at ScienceDirect

Psychiatry Research: Neuroimaging

journal homepage: www.elsevier.com/locate/psychresns

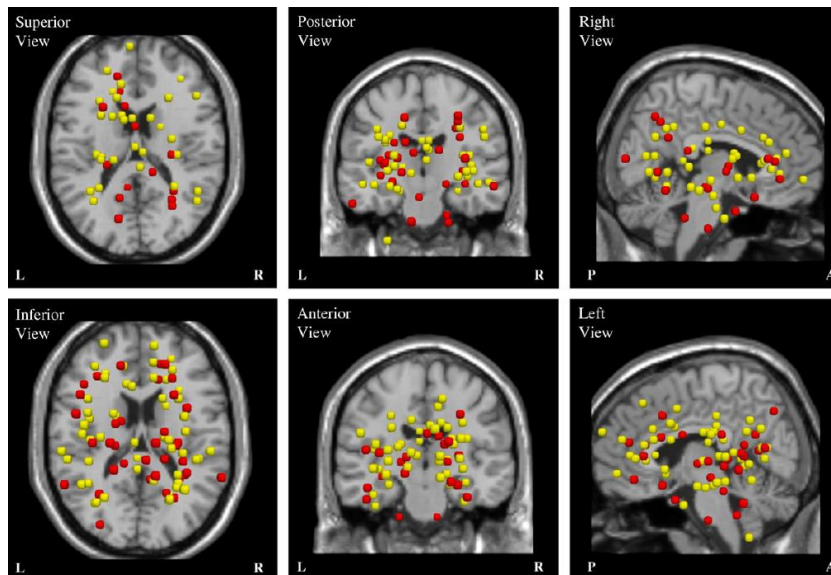


Voxel-based morphometry (VBM) studies in schizophrenia—can white matter changes be reliably detected with VBM?

Eric D. Melonakos^a, Martha E. Shenton^{a,b}, Yogesh Rathi^a, Douglas P. Terry^a, Sylvain Bouix^a, Marek Kubicki^{a,b,*}

^a Psychiatry Neuroimaging Laboratory, Department of Psychiatry, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA

^b Department of Psychiatry, VA Boston Healthcare System, Harvard Medical School, Boston, MA, USA



1. Medline and Ovid

- DTI, Diffusion, VBM, Voxel Based, whole brain y schizophrenia

2. Resultados (23 artículos)

- 13 Chronic schizophrenia
- 10 Early psychosis

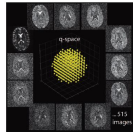
3. White matter coordinates

5. Results

High Dispersion of the results
Maximum: 5 regions appear in the 26 % of the studies

Tract-Based Spatial Statistics (TBSS-VBM)

Diffusion Weighted Images



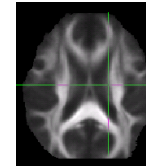
Movement correction

Eddy current correction

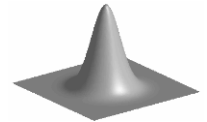
EPI distortion correction

Computing Scalar Maps (MD, FA, etc)

Stereotactic Space



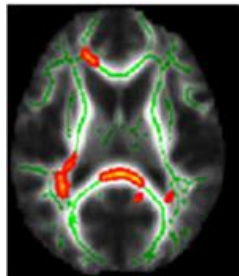
Smoothing Kernel



Smoothing

Individual pre-processing

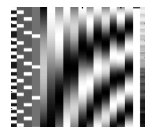
Statistical map



Group-wise processing

Statistical processing

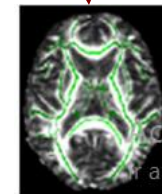
General Linear Model



Design Matrix

Project the FA Over the Skeleton

FA Skeletonization



FA: Fractional Anisotropy

MD: Mean Diffusivity

EPI: Echo Planar Imaging

Fractional anisotropy skeleton: TBSS (FSL)

Projected FA

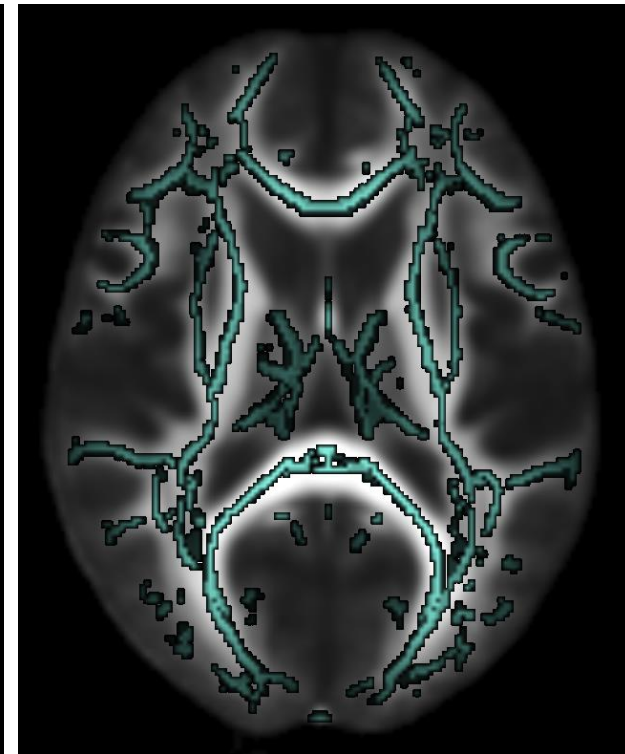
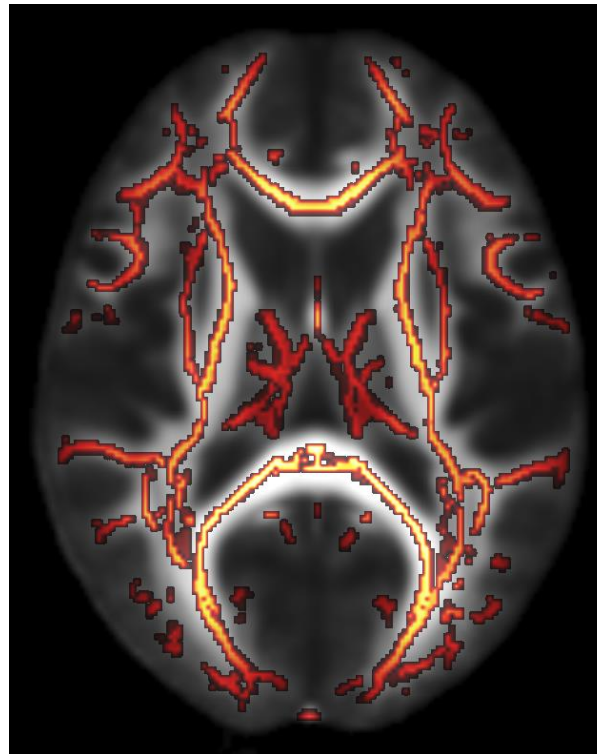
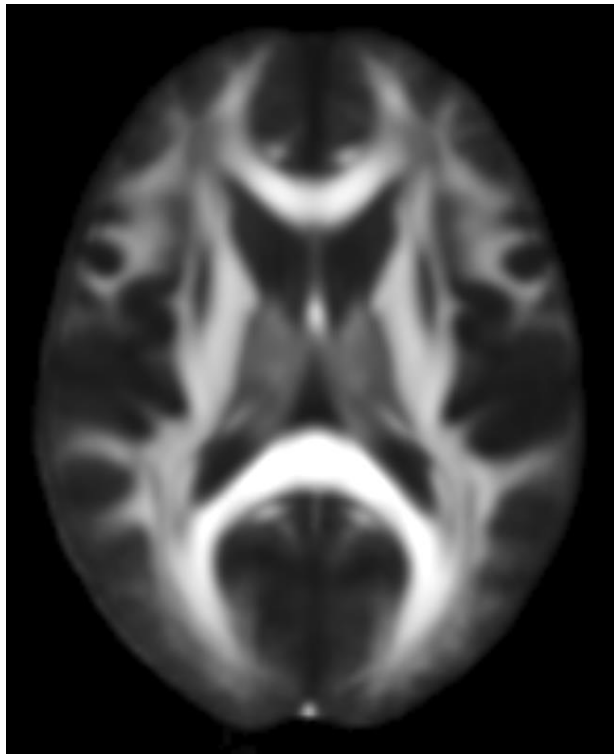
(Thinning algorithm)

+

Mean FA image

(FA over the skeleton)

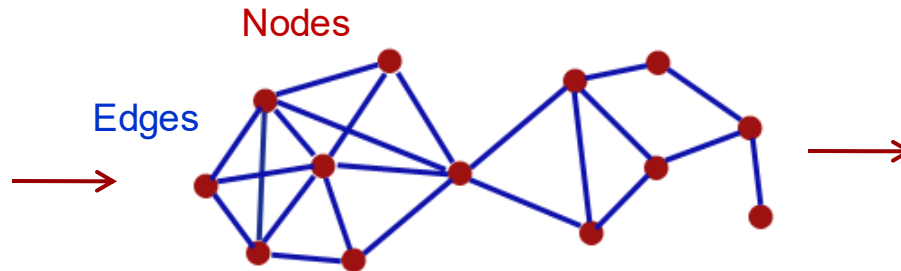
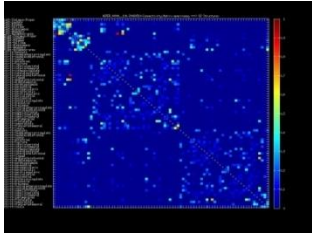
FA Skeleton



Network analysis

■ Graph theory

Connectivity Matrix

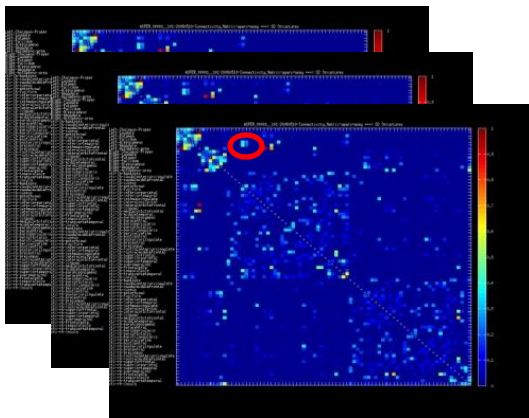


Network properties

- Degree
- Path length
- Shortest path
- etc

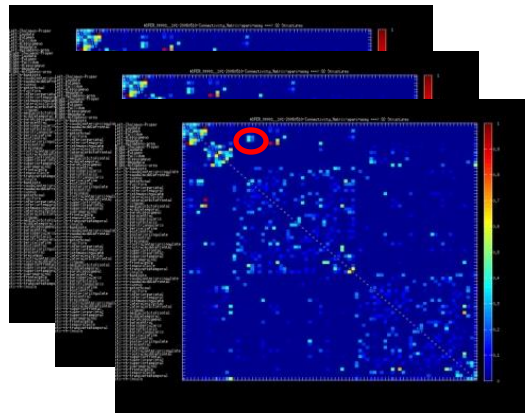
■ Differences between groups for each connection

Group 1



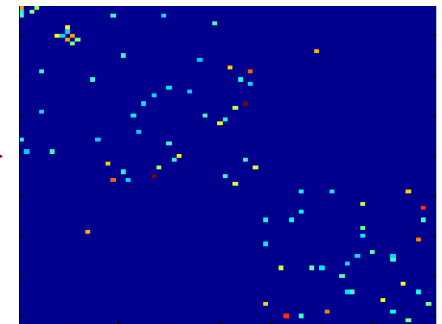
Vs

Group 2



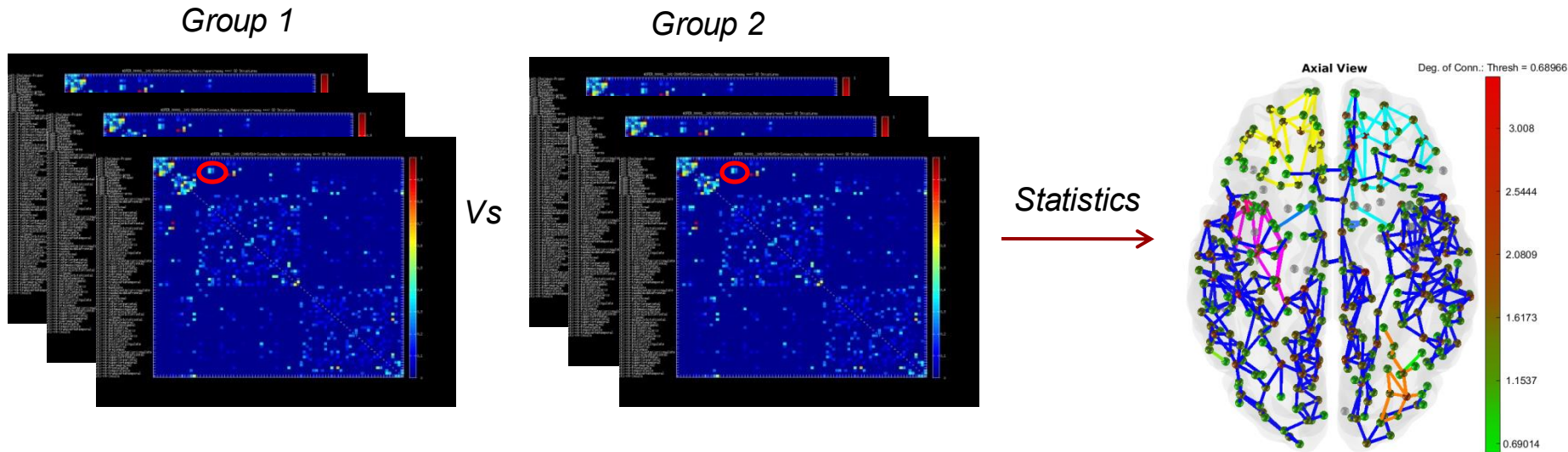
Statistics

$p\text{-value} < 0.05$
+
FDR correction



Network Based Statistics (NBS, Matlab)

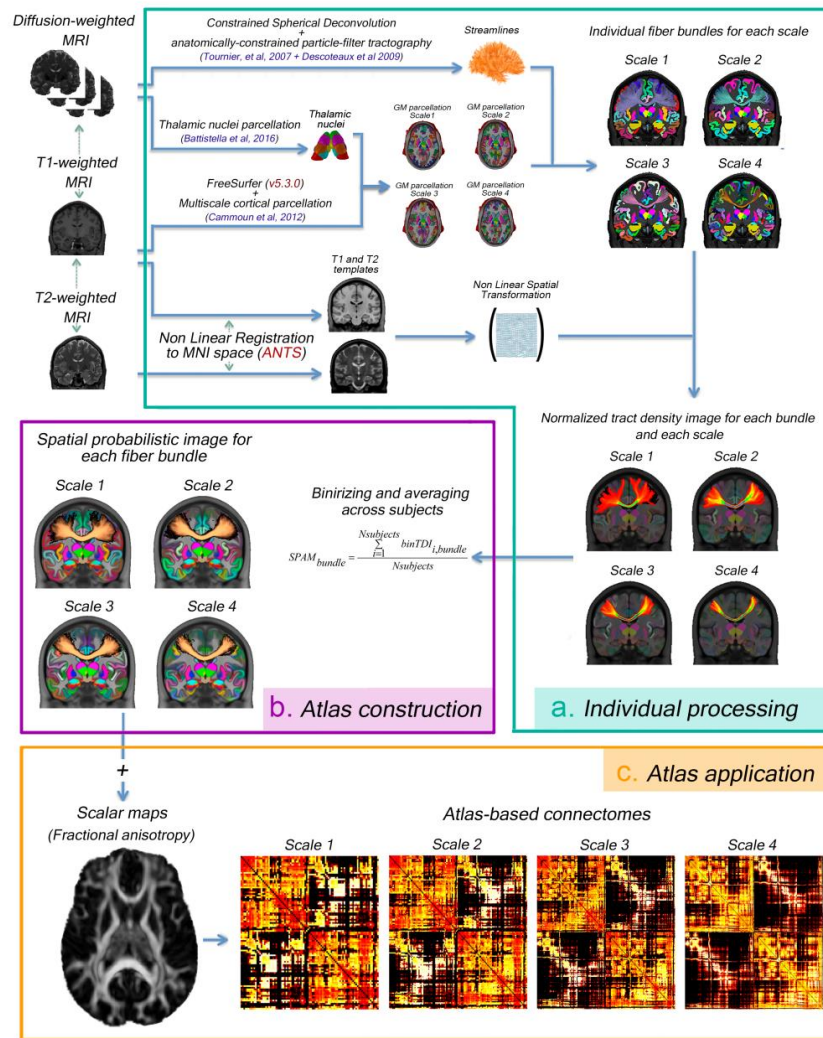
■ Brain Connectivity Toolbox (BCT, Matlab)



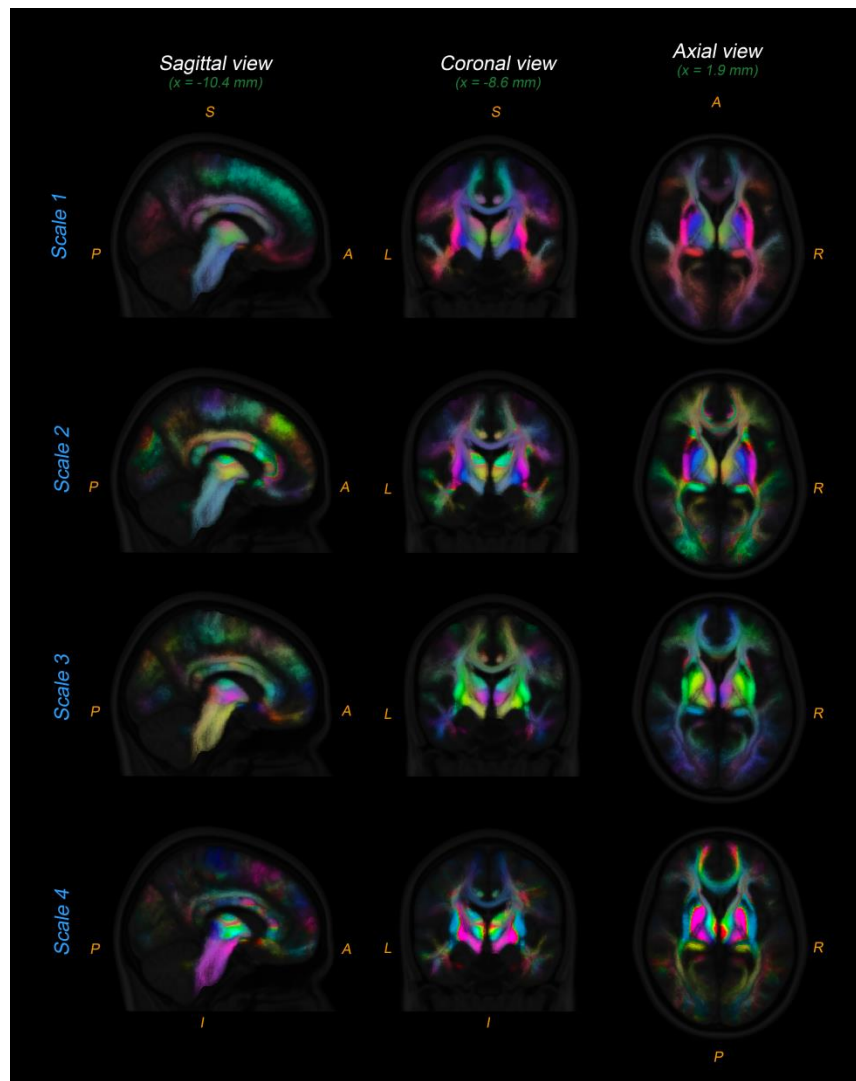
■ NBS main steps

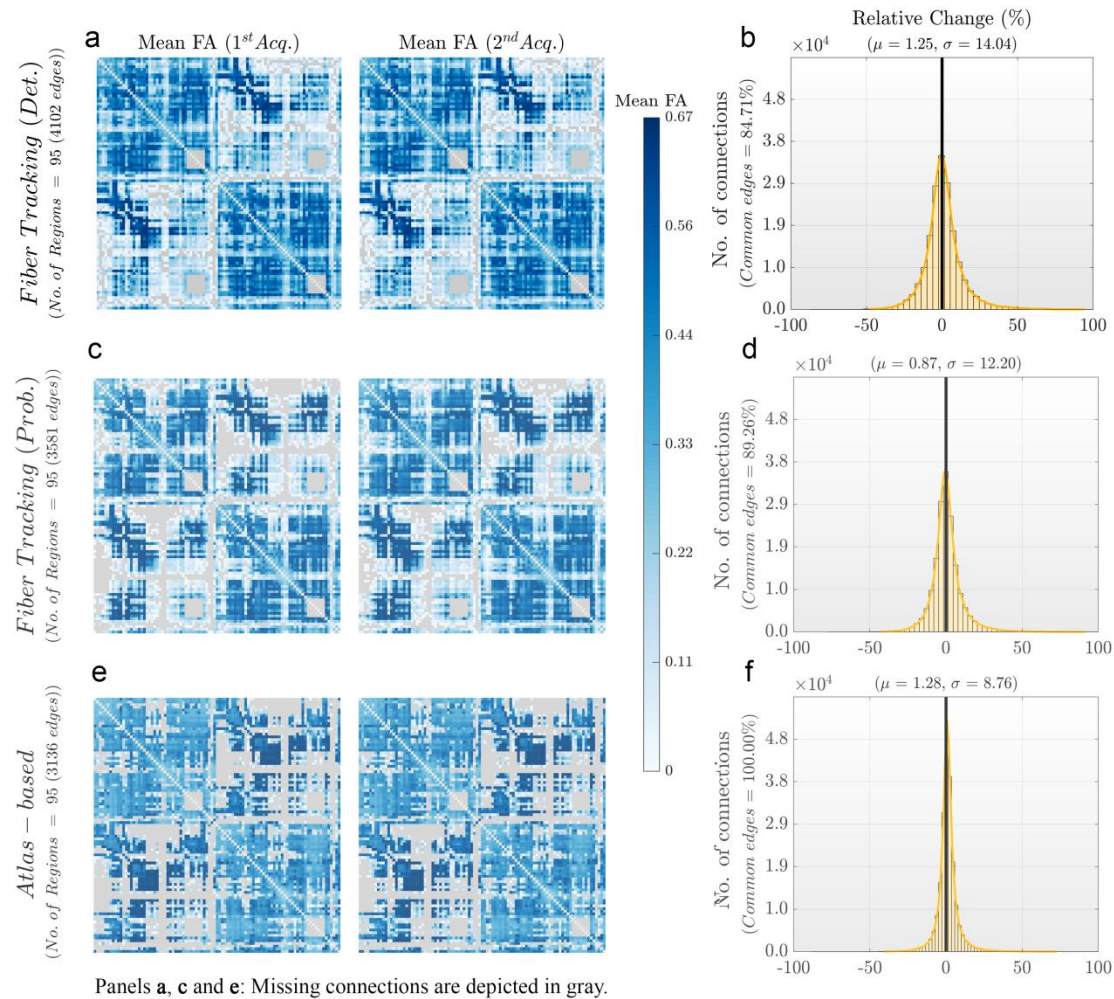
1. Perform a two-sample T-test at each edge independently
2. Threshold the T-statistic available at each edge to form a set of supra-threshold edges.
3. Identify any components in the adjacency matrix defined by the set of supra-threshold edges. Compute the size of each observed component identified (i.e., the number of edges it comprises).
4. Repeat K times steps 1-3, each time randomly permuting members of the two populations and storing the size of the largest component identified for each permutation.

Multi-scale atlas of the human connectome

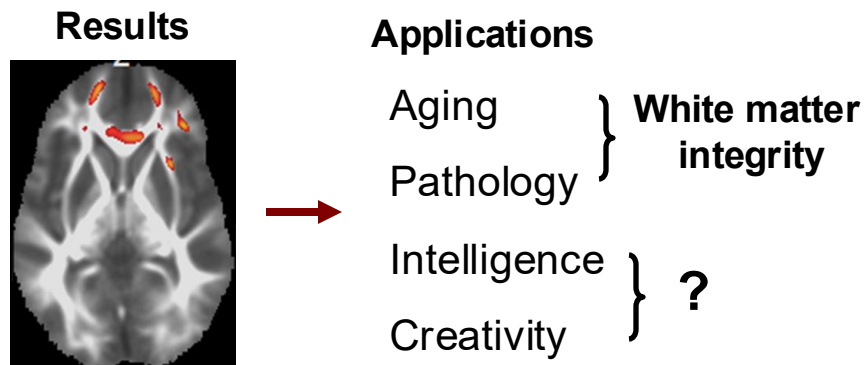
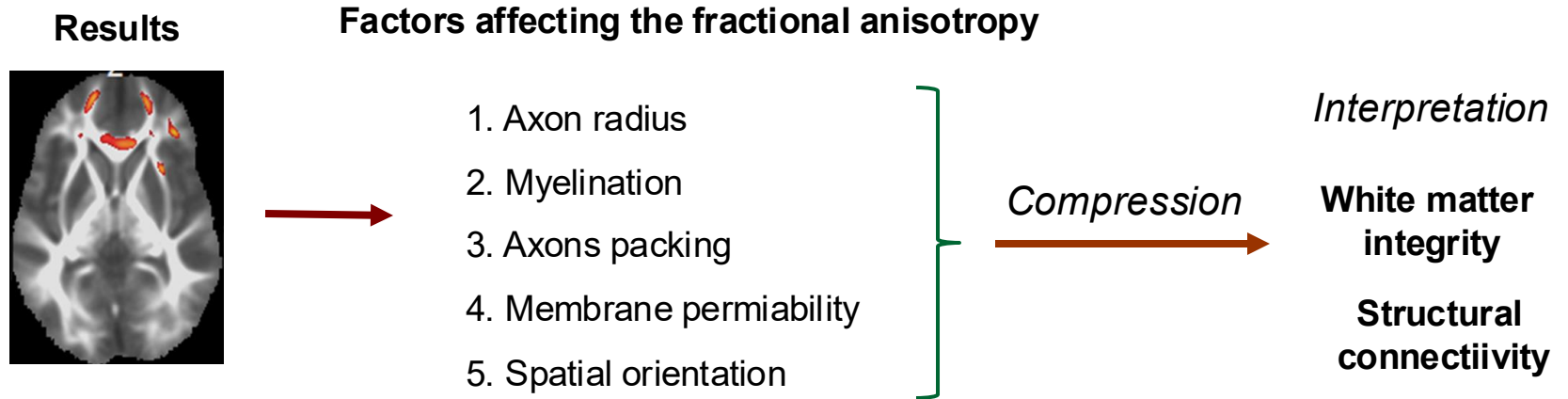


Multi-scale atlas of the human connectome





Interpretation (FA)

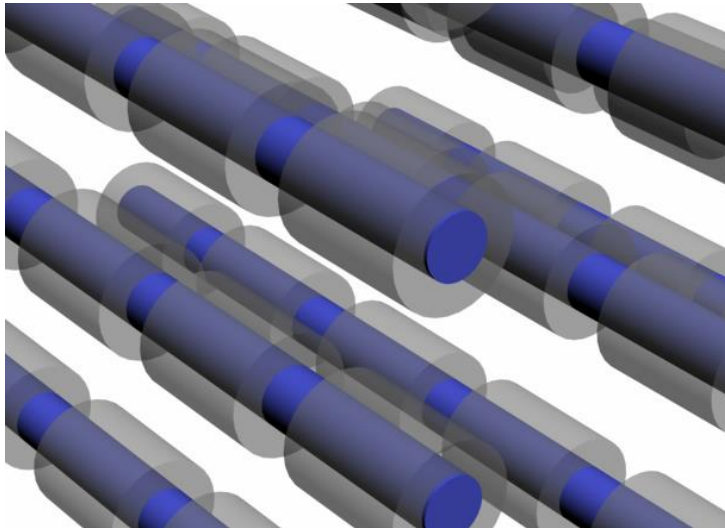


Interpretation of the results (FA)

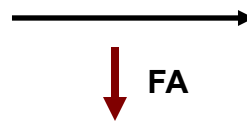
2 Voxels: A and B

$\left. \begin{array}{l} = \text{Axon radius} \\ = \text{Myelination} \\ = \text{Axons packing} \\ = \text{Spatial orientation} \end{array} \right\} = \text{White Matter Integrity}$

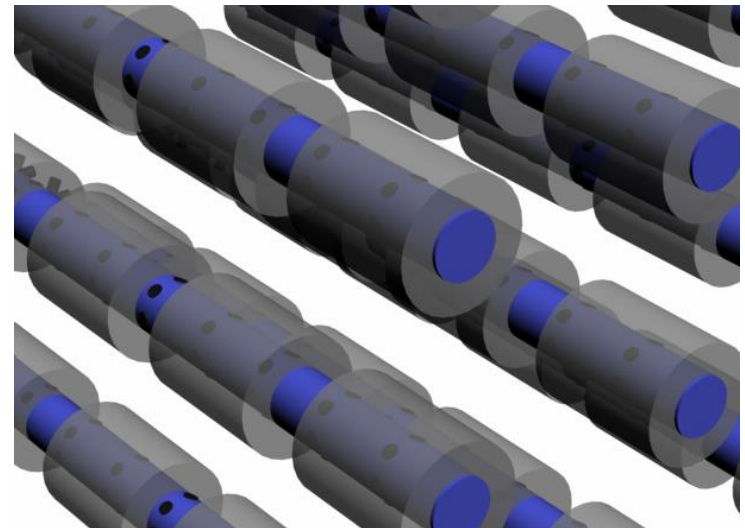
Voxel A



Change in
membrane
permeability



Voxel B



↓ FA is NOT necessarily a ↓ in white matter integrity

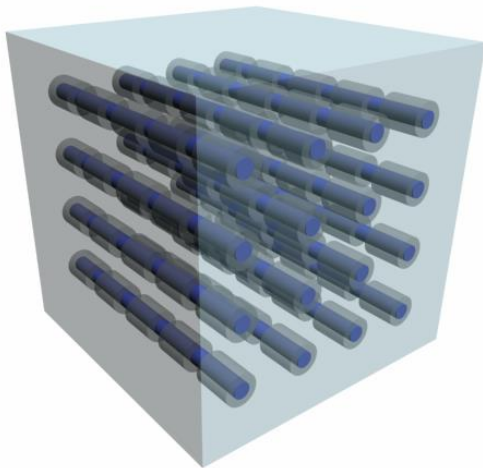
Interpretation of the results (FA)

2 Voxels: A and B

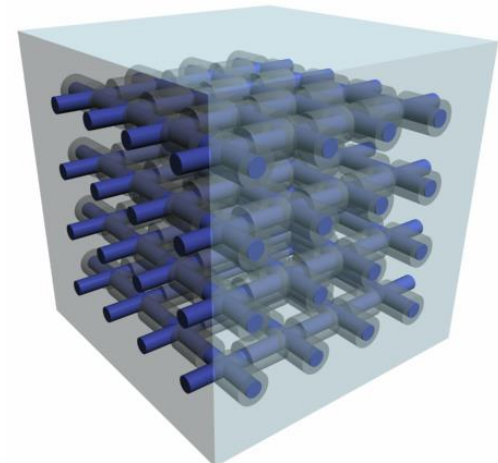
= Axon radius
= Myelination
= Axons packing
= Membrane permeability

} = White Matter Integrity

Voxel A



Voxel B



Fibers crossing

↓ FA

↓ FA is NOT necessarily ↓ White Matter Integrity

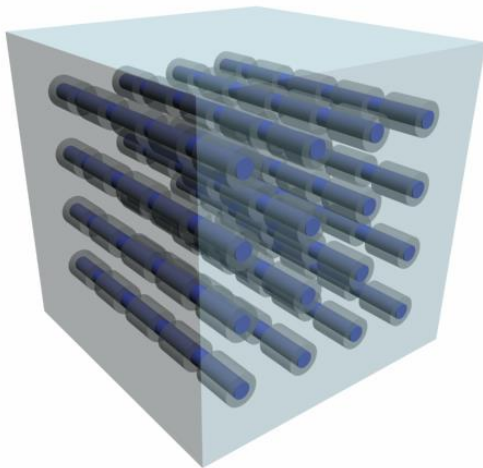
Interpretation (RD)

2 Voxels: A and B (Shiverer mice)

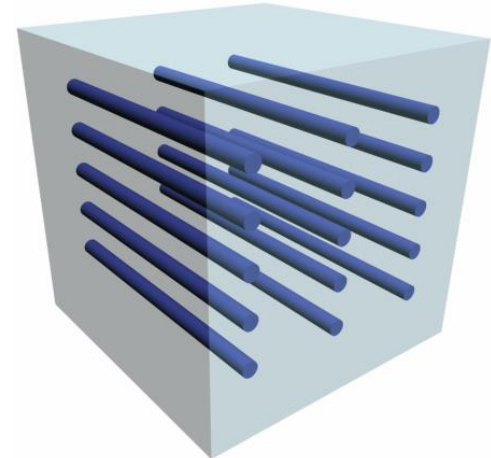
White Matter Integrity

- = Axon radius
- = Spatial orientation
- = Axons packing
- = Membrane permeability

Voxel A



Voxel B



↓ Myelination

↑ RD

↑ RD is NOT necessarily ↓ White Matter Integrity

Interpretation (Connectivity)

2 Configurations: A and B

= Axon radius

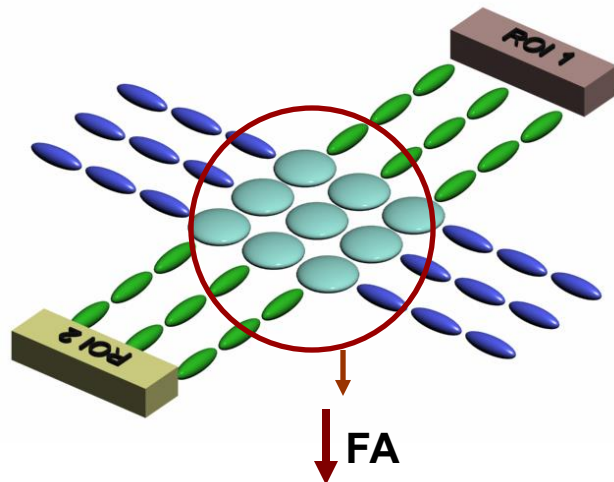
= Myelination

= Axons packing

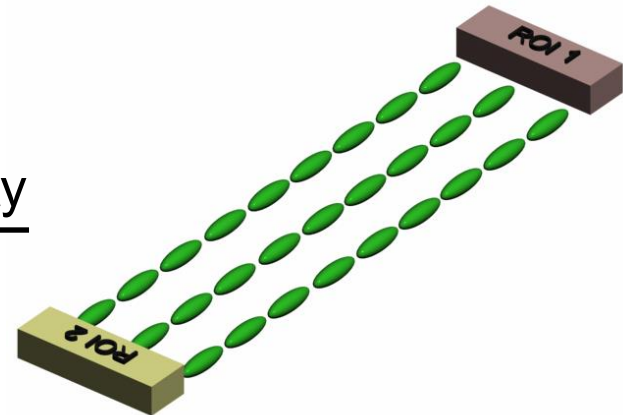
= Membrane permeability

= White Matter Integrity

Configuration A



Configuration B



= Connectivity

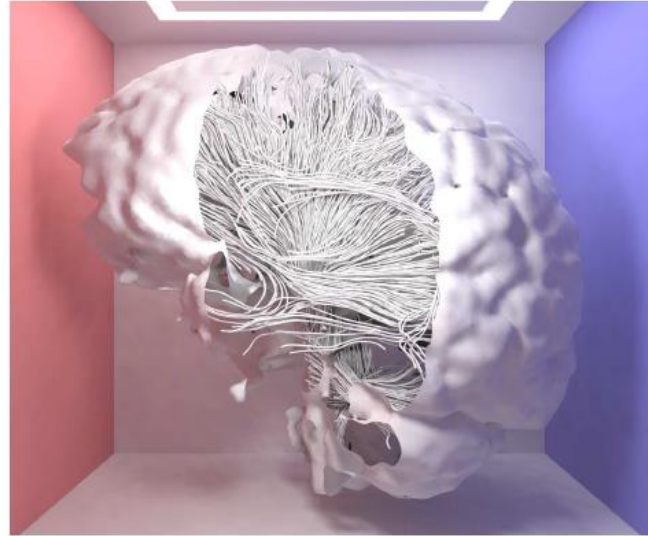
↓ FA is NOT necessarily a ↓ in white matter integrity

List of tools

Software/tools	URL	Main Purpose
<i>3D Slicer</i> (Pieper et al., 2006)	http://www.slicer.org/	Preprocessing and tensor estimation
<i>AFNI</i> (Cox, 2012)	http://afni.nimh.nih.gov/afni	Preprocessing and tensor estimation
<i>Biolmage Suite</i> (Papademetris et al., 2005)	http://www.bioimagesuite.org/	Tensor estimation, ODF estimation, ROI analysis, and tractography
<i>BrainVoyager QX</i> (Goebel, 2012)	http://www.brainvoyager.com/	Tensor estimation and tractography
<i>Camino</i> (Cook et al., 2006)	http://camino.cs.ucl.ac.uk/	Preprocessing, tensor estimation, ODF estimation, ROI analysis, and tractography
<i>Dipy</i> (Garyfallidis et al., 2011)	http://dipy.org	Tensor estimation, ODF estimation, ROI analysis, and tractography
<i>DTI-TK</i> (Mang et al., 2009)	http://dti-tk.sourceforge.net/pmwiki/pmwiki.php	Tensor estimation and tractography
<i>DTIStudio</i> (Jiang et al., 2006)	https://www.mristudio.org/wiki/DtiStudioV2	Tensor estimation, ROI analysis and tractography
SMT (Kaden et al., 2007)	https://github.com/ekaden/smt	Microstructure maps estimation
AMICO (Dadducci et al., 2015)	https://github.com/daducci/AMICO	Microstructure maps estimation
QSIPREP	https://qsiprep.readthedocs.io/en/latest/	Preprocessing, tensor estimation, ODF estimation, ROI analysis, and tractography

List of tools

Software/tools	URL	Main Purpose
<i>ExploreDTI</i> (Leemans et al., 2009)	http://www.exploredti.corri/	Preprocessing, tensor estimation, ROI analysis, and tractography
<i>Freesurfer</i> (Fisch, 2012)	http://surfer.nmr.mgh.harvard.edu/	Preprocessing, tensor estimation, fODF estimation and and tractography
<i>FSLFDT</i> (Smith et al., 2004)	http://www.fmrib.ox.ac.uk/fsl/fdt/index.html	Preprocessing, tensor estimation, fODF estimation and and tractography
<i>FSLTBSS</i> (Smith et al., 2006)	http://www.fmrib.ox.ac.uk/fsl/AbssAndex.html	TBSS analysis
<i>Mrtrix3</i> (Tournier et al., 2012)	http://www.mrtrix.org/	Preprocessing, tensor estimation, fODF estimation and and tractography
<i>SATURN</i> (Cardenes et al., 2010)	http://www.lpi.tel.uva.es/saturn/	ensor estimation and tractography
<i>SPM and toolboxes</i> (e.g., <i>Diffusion DTI Toolboxes</i>)	http://www.fil.ion.ucl.ac.uk/spm/ext/	Preprocessing and tensor estimation
<i>Trackvis and Diffusion Toolkit</i> ,. (Wang et al., 2007)	http://trackvis.org/	Tensor estimation, ODF estimation and tractography
<i>DSI-Studio</i>	http://dsi-studio.labsolver.org/	Preprocessing, tensor estimation, ODF estimation, ROI analysis, and tractography
<i>PHARDI</i> (García-Blas, et al 2018)	https://github.com/arcosuc3m/phardi	ODF estimation



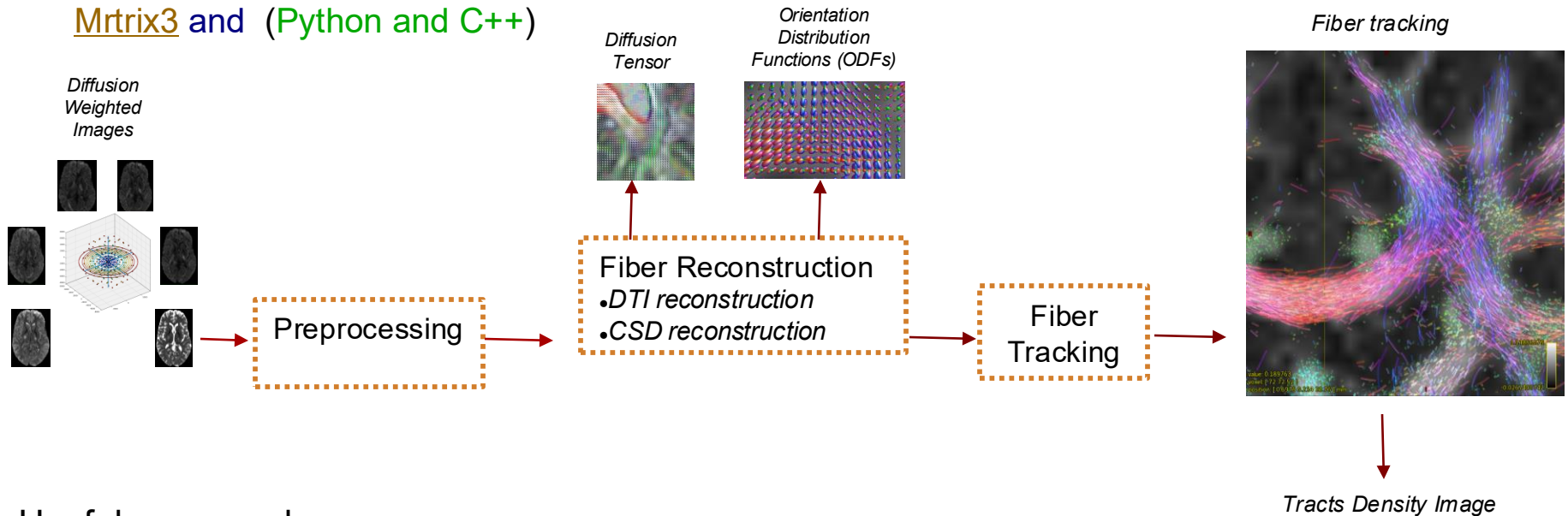
MRI Processing Theory: Diffusion Weighted Imaging

Yasser Alemán-Gómez

Radiology department. CHUV, Switzerland
October 1st, 2026

Implemented pipelines

Mrtrix3 and (Python and C++)

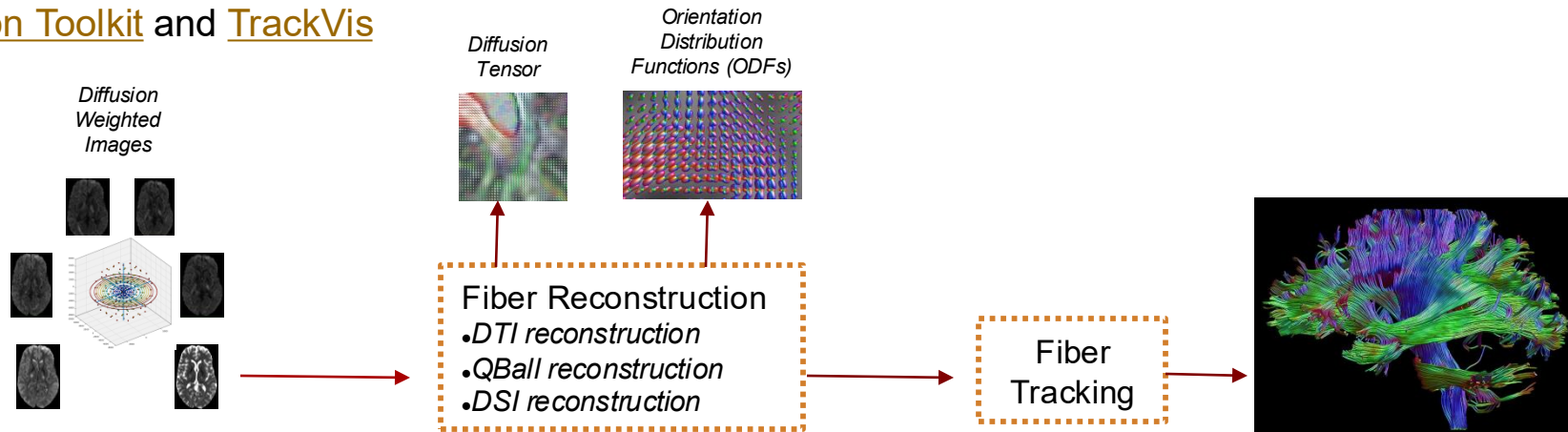


Useful commands

- dwidenoise** (data denoising and noise map estimation)
- dwi2tensor** (diffusion tensor estimation)
- tensor2metric** (generate maps of tensor-derived parameters)
- dwi2response** (intravoxel fiber reconstruction, bayesian framework)
- dwi2fod** (fiber ODFs estimation using CSD)
- tckgen** (probabilistic or deterministic tracking)
- tckmap** (compute tracts density image)

Implemented pipelines

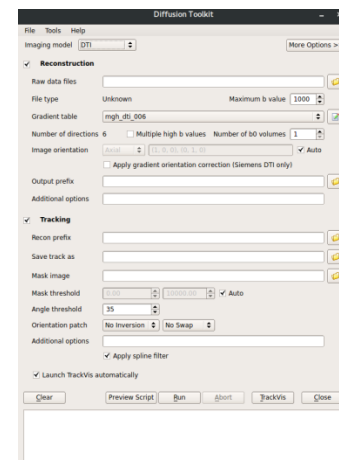
Diffusion Toolkit and TrackVis



Useful commands

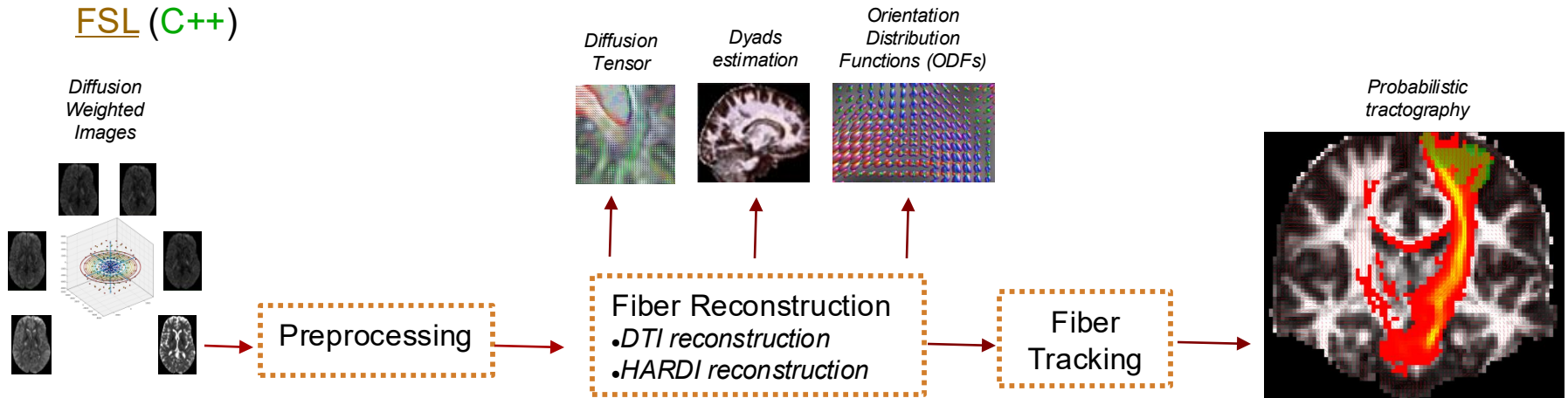
- dti_recon** (for diffusion tensor reconstruction)
- odf_recon** (for ODFs reconstruction)
- dti_tracker** (fiber tracking using the diffusion tensor)
- odf_tracker** (fiber tracking using the ODFs)
- dtk** (lunch the diffusion toolkit graphical interface)
- trackvis** (lunch the trackvis graphical interface)

Graphical User Interface



Implemented pipelines

FSL (C++)



Graphical User Interface

Useful commands

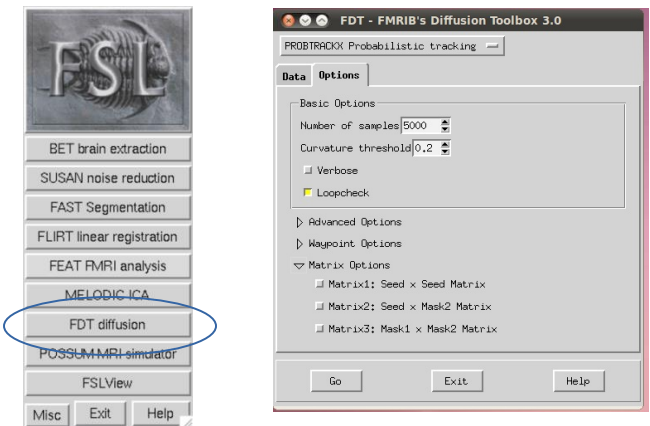
eddy (motion, eddy current and distortions correction)

dtifit (tensor fitting)

bedpostx (intravoxel fiber reconstruction, bayesian framework)

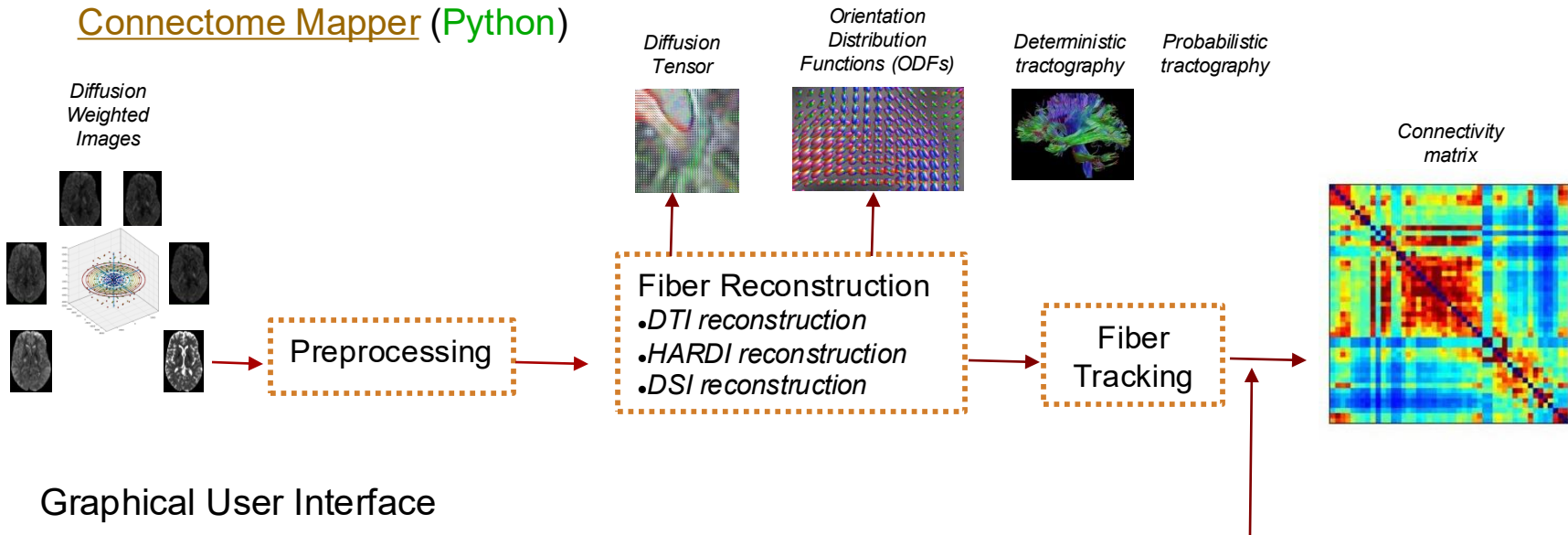
qbboot (ODFs reconstruction using CSA model)

probtrackx (probabilistic tracking, bayesian framework)

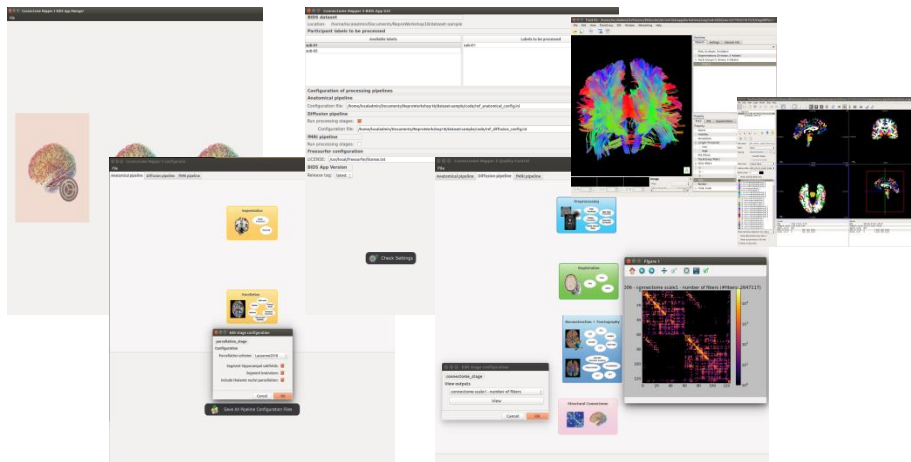


Implemented pipelines

Connectome Mapper (Python)



Graphical User Interface



Anatomical parcellation

